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Chemical Kinetics

It deals with measurement of reaction rates, mechanism, the factors which affect the rate of a chemical reaction.

Rate of chemical reaction is rate of production of chemicals per unit time, & it is really important

(B) Important points of a reaction →

THERMO 1.) Spontaniety (whether it is ~~feasible~~ or not)

EQUILIB 2.) Extent of reaction

3.) Rate of reaction (CHEMICAL KINETICS)

→ Thermodynamics : macroscopic study (of bulk matter)

→ Chemical Kinetics : microscopic study (or molecular study) of matter.

Rate of Disappearence & rate of appearence =>

eg → $2A + B \rightarrow C + 3D$

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P.T.O.

• Rate of disappearance is defined for reactants as they are being used.

• Rate of appearence is defined for products as they are formed.

for :- $2A + B \rightarrow C + 3D$

Rate of Disappearence of reactant 'A' (or rate of reaction of A) $\Rightarrow$

$$= \frac{\text{change in conc. of A}}{\text{Time taken}}$$

$$= -\frac{\Delta[A]}{\Delta t}$$ { Average rate of Disappearence of A }

$$= -\frac{d[A]}{dt}$$ { Instaneous rate of disappearence of A }

Rate of rx^n of B (or rate of disappearence of B) $\rightarrow$

$$= -\frac{\Delta[B]}{\Delta t}$$

$$\Rightarrow = -\frac{d[B]}{dt}$$

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- Rate of disappearence is defined for reactants as they are being used.
- Rate of appearence is defined for products as they are formed.

for :- $2A + B \rightarrow C + 3D$

Rate of Disappearence of reactant 'A' (or rate of reaction of A) $\Rightarrow$

$$= \frac{\text{change in conc. of A}}{\text{time taken}}$$

$$= -\frac{\Delta[A]}{\Delta t}$$ { Average rate of Disappearence of A }

$$= -\frac{d[A]}{dt}$$ { Instaneous rate of disappearence of A }

Rate of rxn of B (or rate of disappearence of B) $\rightarrow$

$\Delta[B] = [B]_{t_2} - [B]_{t_1}$
$= -ve$

$$= -\frac{\Delta[B]}{\Delta t}$$ average rate of disappearance of B

$$\Rightarrow = \frac{d[B]}{dt}$$ instantaneous rate of disappearance of B

[ ] : denotes molar conc.

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Rate of appearence of C = or rate of reaction of C

$$\Rightarrow + \frac{\Delta[C]}{\Delta t}$$ (avg: rate of appearence of C)

$$\Rightarrow + \frac{d[C]}{dt}$$ (instaneous rate of appearence of C)

(+) Products ke rate of rxn ke liye '+' sign lagate hai, while reactants ke liye '-'.

Rate of appearence of D ⇒

$$= + \frac{\Delta[D]}{\Delta t}$$

$$= \frac{d[D]}{dt}$$

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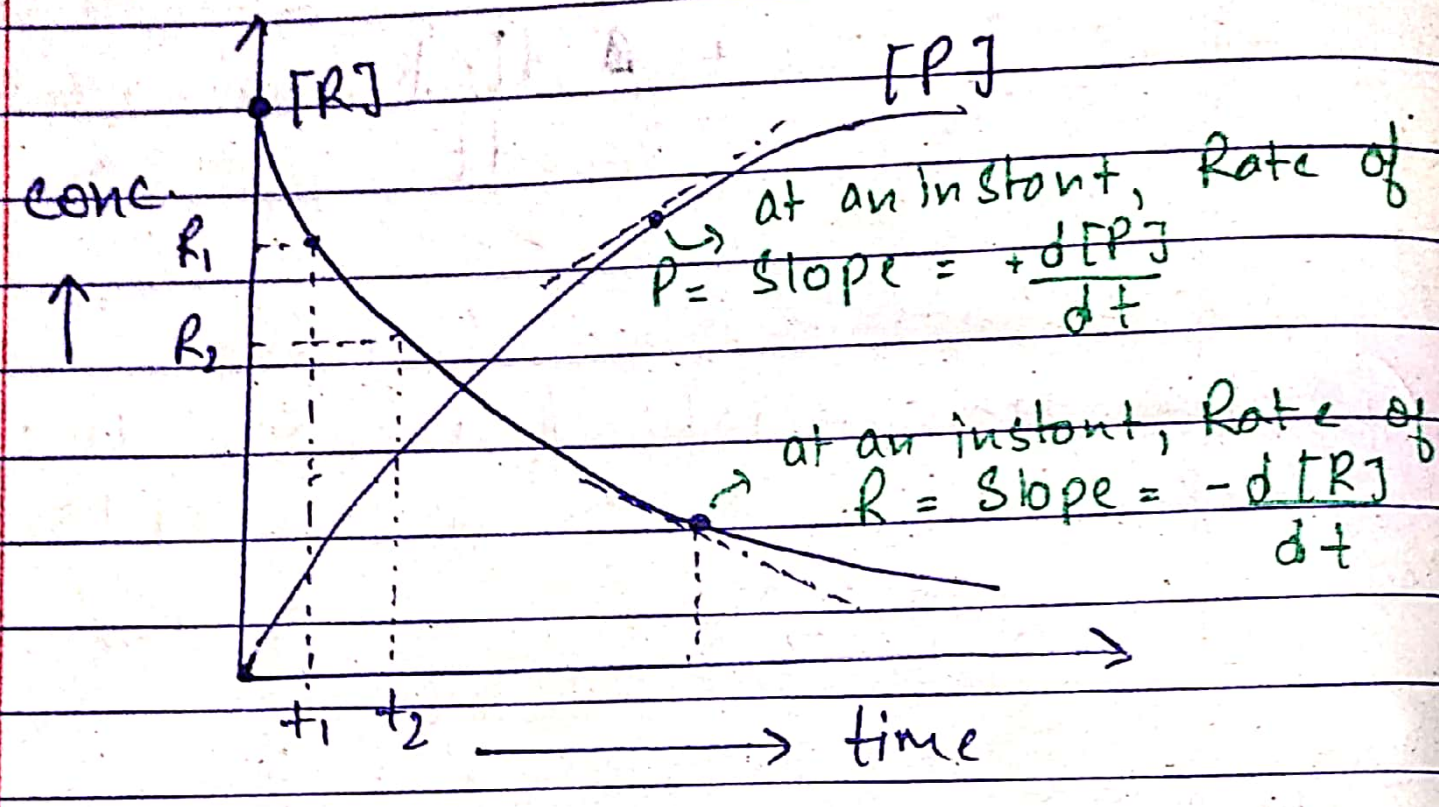
Graphically,

Reactant (R) ⟶ Product (P)



(*) Rate of disappearence of R

$$= -\frac{\Delta[R]}{\Delta t}$$

$$= -\left(\frac{R_2 - R_1}{t_2 - t_1}\right)$$

(*) Unit of Rate of reaction of Reactants | Products

⇒ mol | lit | time

or

⇒ atm | time or Pa | time

(for gases)

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Relation b/w rate of Appearence & Rate Of Disappearence ⇒

eg → $2A + B \rightarrow C + 3D$ (Chemical Rxn, Balanced)

- Upar wali rxn me, B ke har ek mole ke liye, A ke do moles kharch ho rahe hai.

Hence, Rate of disappearence of A = 2 × Rate of disappearence of B

$$\Rightarrow -\frac{d[A]}{dt} = \left\{-\frac{d[B]}{dt}\right\} \times 2$$

Similarly ⇒

$$\Rightarrow \frac{d[C]}{dt} = -\frac{d[B]}{dt}$$

$$\Rightarrow -\frac{d[D]}{dt} = 3 \times \left\{-\frac{d[B]}{dt}\right\}$$

$$\Rightarrow \boxed{-\frac{1}{2}\frac{d[A]}{dt} = -\frac{1}{1}\frac{d[B]}{dt} = +\frac{1}{1}\frac{d[C]}{dt} = \frac{1}{3}\frac{d[D]}{dt}}$$

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In general,

$$aA + bB \rightarrow cC + dD$$

$$\Rightarrow \boxed{\frac{-1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = \frac{1}{c}\frac{d[C]}{dt} = \frac{1}{d}\frac{d[D]}{dt}}$$

NOTE

Above relation is only applicable when all the reactants & products are in same phase.

same phase

Rate of Reaction ⇒

For a given reaction, rate of reaction is defined as the rate of change of conc. of any substance involved in reaction divided by its stoichometric coffecient.

eg → $aA_{(aq)} + bB_{(aq)} \rightarrow cC_{(aq)} + dD_{(aq)}$

Rate of reaction $= -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt}$

$= \frac{1}{c}\frac{d[C]}{dt} = \frac{1}{d}\frac{d[D]}{dt}$

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Units → mol / lit / time

or

M. time^{-1}

or

Pa or atm. time^{-1}

NOTE

If Stoichometry of given rxn is altered by multiplying by a factor, then note that rate of disappearence of reactants & rate of appearence of products do not change, however rate of reaction will be change.

eg → $A + 2B \longrightarrow 3C$ → Rate of Reaction → 2×10^{-3} M min^{-1}

Rate of Disappearance $\left\{ -\frac{d[A]}{dt} = 2 \times 10^{-3} \text{ M min}^{-1} \right.$

for → $2A + 4B \rightarrow 6C$

↳ for this rxn

Rate of Disappearance $\left\{ = -\frac{d[A]}{dt} = 2 \times 10^{-3} \text{ M min}^{-1} \right.$

Rate of rxn → 1×10^{-3} M min^{-1}

Q: Heating of $(CH_3)_2N_2(g)$ gives $N_2(g)$ & $C_2H_6(g)$. If rate of formation of N_2 is 56 g/lit/sec then find, rate of formation of C_2H_6.

Sol →

$$(CH_3)_2N_2(g) \rightarrow \underset{(g)}{C_2H_6} + \underset{(g)}{N_2}$$

$$\frac{d[N_2]}{dt} = \frac{56}{28} = 2\,M\,sec^{-1}$$

$$\frac{d[C_2H_6]}{dt} = 2M/L/sec$$

as stoichometric coeff. of C_2H_6 & N_2 is 80M.

$$\frac{d[C_2H_6]}{dt} = 2M/L/sec$$

$$= (2\times 30)\,g/L/sec$$

$$= 60\,g/L/sec$$

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Q. $3A_{(g)} \rightarrow 4B_{(g)} + 2C_{(g)}$

↳ Rxn is carried out in a closed vessel of 10L capacity, no. of moles of B increases by 8×10^{-3} moles in 10sec, find →

i.) Rate of Disappearence of A
ii.) Rate of formation of C
iii.) Rate of reaction

Soln:

$$-\frac{1}{3}\frac{d[A]}{dt} = \frac{1}{4}\frac{d[B]}{dt} = \frac{1}{2}\frac{d[C]}{dt}$$

$$\Rightarrow \frac{d[B]}{dt} = \frac{(8\times10^{-3})}{10\times10}$$

$$\Rightarrow \underline{8\times10^{-5}\ M\ sec^{-1}}$$

(i.) $-\frac{d[A]}{dt} = \frac{3}{4}(8\times10^{-5}) = 6\times10^{-5}\ M\ sec^{-1}$

(ii.) $\frac{d[C]}{dt} = \frac{2}{4}(8\times10^{-5}) = 4\times10^{-5}\ M\ sec^{-1}$

(iii.) Rate of rxn $= \frac{1}{2}\frac{d[C]}{dt} = 2\times10^{-5}\ M\ sec^{-1}$

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Factors Affecting Rate of Reaction ⇒

(1) Conc. of reactant / Partial pressure of reactant
(2) Temperature
(3) Catalyst's presence
(4) Surface area

→ main factors that affect rate of rxn

(1) Effect of Conc. / Partial pressure of Reactant →

Consider a rxn, $aA + bB \xrightarrow{K}$ Product

Experimentally,

Rate of reaction $\propto [A]^p [B]^q$

$$\text{Rate of rxn} = K[A]^p[B]^q$$

Experimental Rate Law

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⇒ Rate law is the expression which relates rate of rxn & conc. of reactants

p = partial orders of rxn wrt A

q = partial orders of rxn wrt B

→ n = (p+q) = order of rxn

K = Rate const. of reaction or specific reaction rate

Order of Reaction : It is the sum of exponents of all concentration terms appearing in rate law. It can be fractional, integer, +ve or -ve, or zero.

NOTE: Order reflects the sensitivity of rate wrt variation in conc. of reactions.

eg → Rate = K(conc)^order

If order = 3
- conc → 2 times
- rate → 8 times

If order = 0.1
- conc = 2 times
- Rate → almost unaffected

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→ Order cannot be predicted by seeing Stoichometry of rxn. Order is always experimentally measured.

i.e., P, Q need not be equal to the stoichometry coeff. (a, b)

↓
partial orders

eg → $2N_2O_5 \xrightarrow{K} 4NO_2 + O_2$

↳ Rate = $K[N_2O_5]$

(First order rxn)

⇒ Order = 1

$H_2(g) + Cl_2(g) \xrightarrow{h\nu} 2HCl$

↳ Rate = $K[H_2]^0[Cl_2]^0$

= K {constant}

(Zero Order Rxn)

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eg:- $2HI_{(g)} \xrightarrow{K} H_{2(g)} + I_{2(g)}$

$$\text{Rate} = K[HI]^2$$

(2nd order rxn)

$2HI \xrightarrow{\text{on gold surface}} H_{2(g)} + I_{2(g)}$

$$\text{Rate} = K[HI]^0$$

$$= K \{\text{constant}\}$$

(Zero order reaction)

"Remember"

NOTE

→ In general, gaseous decomposition reactions occuring on metal surface with high conc. of gas is zero order rxn.

(eg → decomposition of HI, NH_3 provided that conc. of gas is high)

① ORDER IS ALWAYS EXPERIMENTALLY DEVERMINED.

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Imp NOTE

However, for an elementary rxn (rxn which complete in single step), order of rxn is equal to sum of stoichometric coeff. of reactants.

eg→ $A + 2B \rightarrow$ Products (Elementary rxn)

Rate law :

~~Rate~~ $= K[A]^1[B]^2$

⇒ order of rxn $= 1+2$
$= 3$

(Third order rxn)

→ Rate constant of reaction K :

$$\text{Rate} = K(\text{conc.})^{\text{order}}$$

$$K = \frac{\text{Rate}}{(\text{conc})^{\text{order}}}$$

If conc. of all species in rate law = 1M.

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then || K = Rate ||

⇒ Rate const. of rxn (or specific reaction rate) = rate of reaction when conc. of all reactants involved in rate law are set equal to unity (1)

(a) Unit of K ⇒

$$\frac{\text{mol / lit / time}}{(\text{mol / lit})^{n}}$$

$$\Rightarrow (\text{mol / lit})^{1-n}\ \text{time}^{-1}$$

Unit of K for zero order
= mol / lit / time

" 1st order = time^{-1}

" 2nd order = lit / mol time

→ Rate const. of rxn doesn't depend on conc. or partial pressure.
It depends on temp. & presence of catalyst.

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Note

$$aA + bB \xrightarrow{K} cC + dD$$

{ K → rate const. of rxn }

$$\frac{-1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = \frac{1}{c}\frac{d[C]}{dt} = \frac{1}{d}\frac{d[D]}{dt}$$

$$= \text{Rate of reaction}$$

$$= K[A]^p[B]^q$$

Rate of reaction of A ⇒

Rate of disappearance of A ⇒ $-\frac{d[A]}{dt} = K_A[A]^p[B]^q$

$= (aK)[A]^p[B]^q$

YHP

$$\boxed{\frac{K_A}{a} = \frac{K_B}{b} = \frac{K_C}{c} = \frac{K_D}{d}} \equiv K$$

↳ where K_A, K_B, K_C, K_D are rate const. of A, B, C & D respectively

(*) Jab koi upar generally rate constant kaha hai, unless specified.

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Zero Order Kinetics (or Zero order reactions) ⇒

Differential Rate law

eg → $A \xrightarrow{K}$ Product, $\boxed{\frac{-d[A]}{dt} = K[A]_t^0}$

$= K$ (const.)

let at $t=0$, conc. of $A = a$
& at $t = t$, conc. of $A = (a-x)$

$[A]_0 = a$ (int. conc. of A)
$[A]_t = (a-x)$ (conc. of A at time t)

$$\frac{-d[A]_t}{dt} = K[A]_t^0 = K$$

$$\Rightarrow -\int_{[A]_0}^{[A]_t} \frac{d[A]_t}{dt} = \int_0^t K\,dt$$

$$\Rightarrow -[[A]_t]_{[A]_0}^{[A]_t} = -Kt$$

$$\Rightarrow [A]_t - [A]_0 = -Kt$$

Integrated Rate law ⇒ $\boxed{[A]_t = [A]_0 - Kt}$ → characteristic eqn of zero Order

$$\Rightarrow (a-x) = a - Kt$$

$$\Rightarrow x = Kt$$ } conc. of reactant spent in time 't'

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Note

Simple case of zero order ⇒

eg → $A \xrightarrow{K}$ Product

$$-\frac{d[A]}{dt} = K[A]^0 = K$$

① only 1 reactant is present

② **When two reactants are present ⇒**

$$A + B \xrightarrow{K} \text{Products}$$

$$\text{Rate} = -\frac{d[A]_t}{dt} = -\frac{d[B]_t}{dt} = K[A]^0[B]^0$$

or

$K[A]^1[B]^{-1}$

or

$K[A]^2[B]^{-2}$

Order of all rx^n's = 0

Hence, zero order kinetics

They all are still zero order reactions (zero order kinetics)

③ **Half life ($t_{1/2}$)** = It is the time period in which half of reactant has reacted

for zero order, $[A]_t = [A]_0 - Kt$

when $t = t_{1/2} \Rightarrow [A]_t = \frac{[A]_0}{2}$

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$$\Rightarrow \frac{[A]_0}{2} = [A]_0 - kt_{1/2}$$

$$\Rightarrow (t_{1/2})_{\text{zero-order}} = \frac{[A]_0}{2K}$$

Half life for zero order Kinetics ⇐

$$\Longleftrightarrow (t_{1/2})_{\text{zero-order}} \propto [A]_0$$

(4) Time of Completion (t_{comp}) : Time in which reactant is completely consumed.

when $t = t_{comp}$, $[A]_t = 0$

$$\hookrightarrow 0 = [A]_0 - K \times t_{comp}$$

$$t_{comp} = \frac{[A]_0}{K} = 2 \times t_{1/2}$$

for zero-order reactions ⇐

(5) Zero-order rx^{n}s complete in a definite time-period.

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Characteristics of zero-order reactions ⇒

(1) Since rate of reaction is constant, equal conc. of reactant reacts in equal time period. (Agar 1 mol 10 min me use hue, toh agle 10 min me bhi, 1 mol hi use hoga)

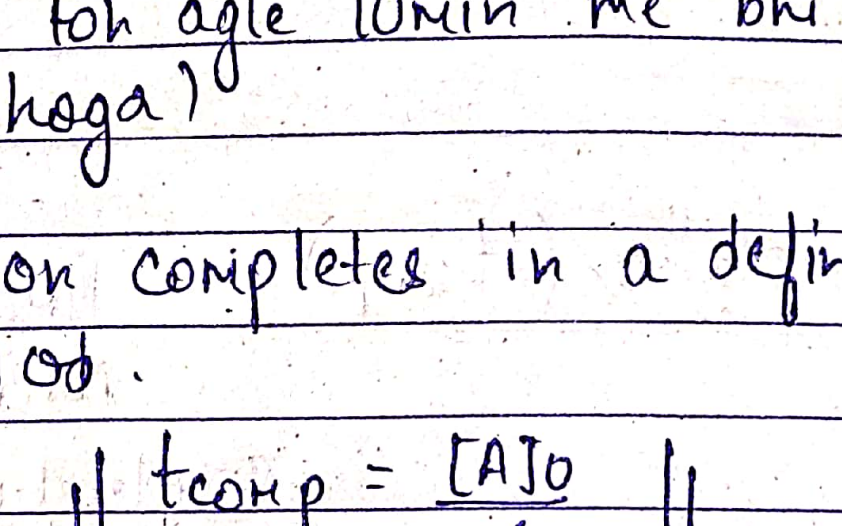
(2) Reaction completes in a definite time period.

$$t_{comp} = \frac{[A]_0}{K}$$

(3) Graphs of zero-order →

$$\text{Rate} = -\frac{d[A]_t}{dt}$$

a.)



$$\text{Rate} = -\frac{d[A]_t}{dt} = K$$

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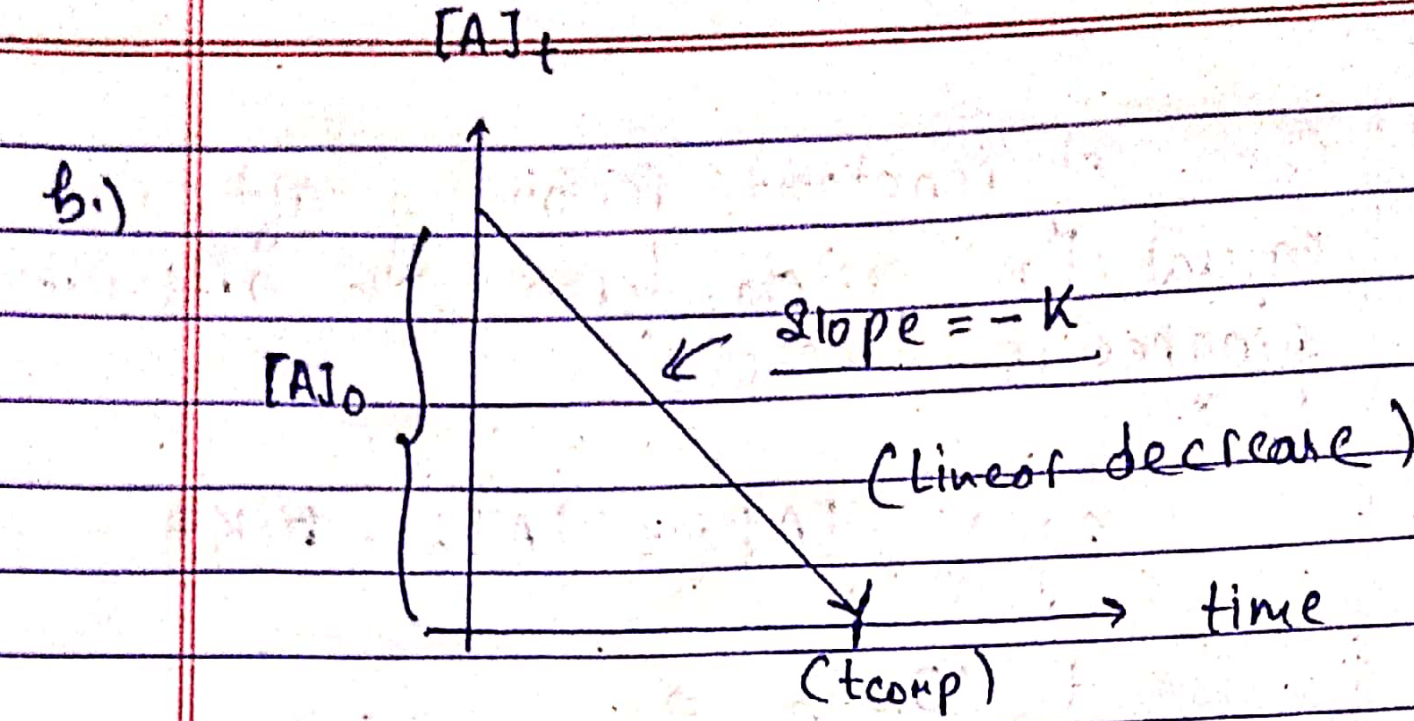
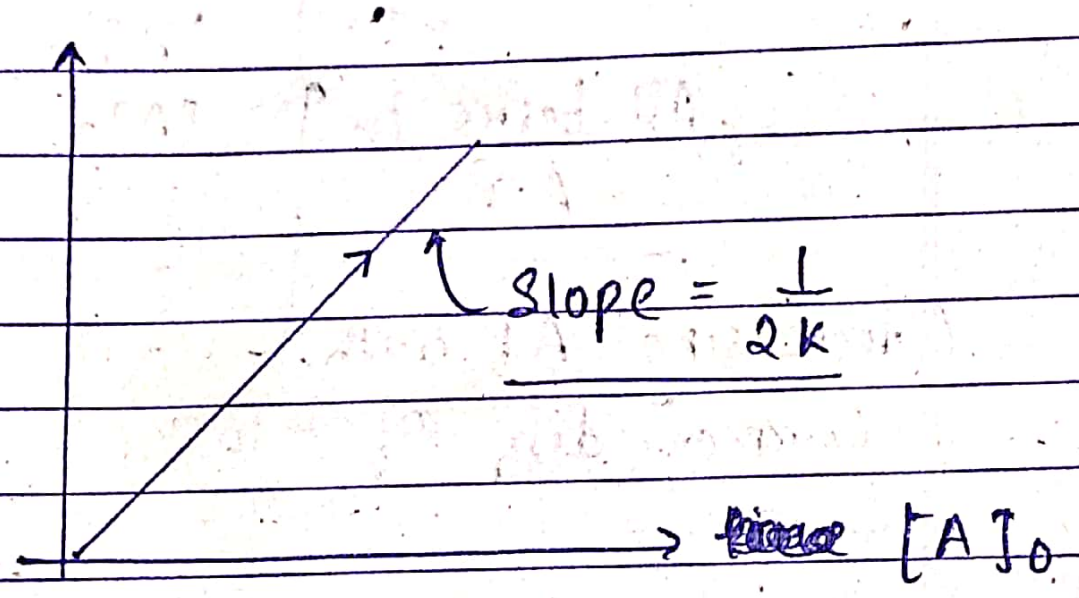
b.)



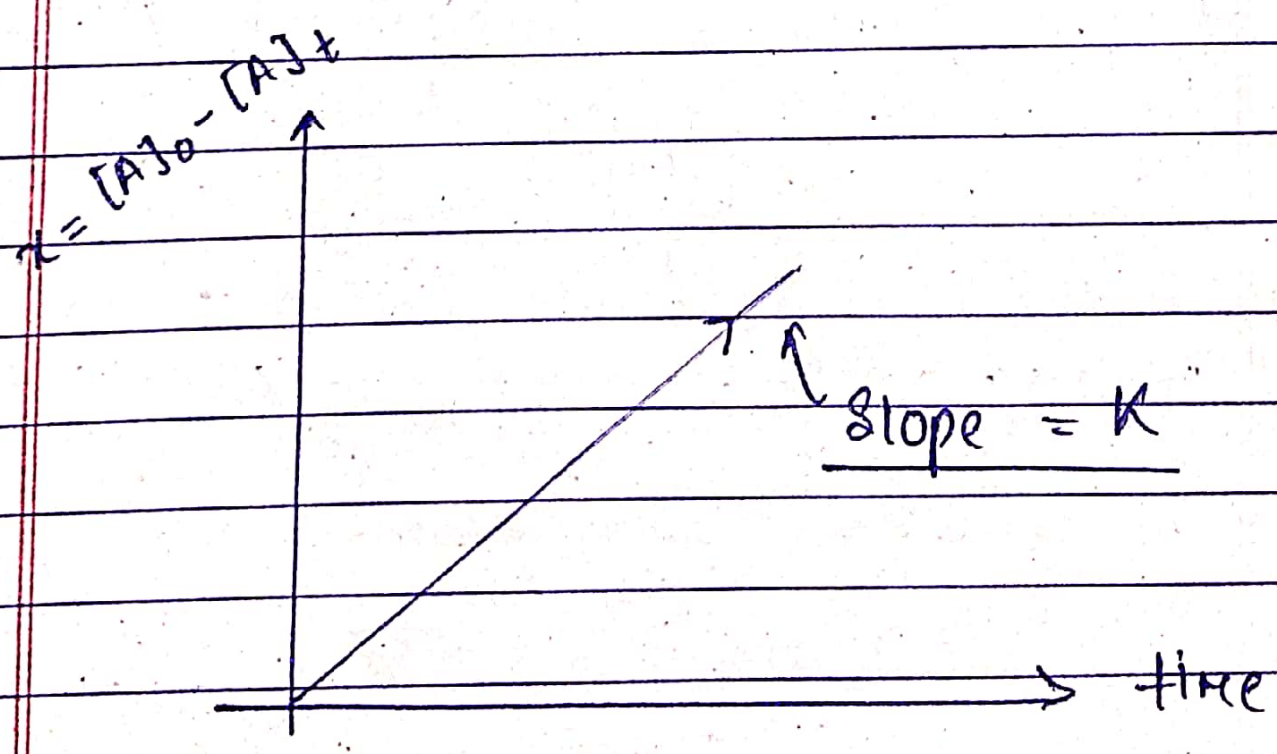
$$[A]t = [A]_0 - Kt$$

c.)



$$t_{1/2} = \frac{[A]_0}{2K}$$

d.)



$$x = Kt$$

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(4) Conc. of reactant remaining after equal time period form an arithmetic progression series.

eg → $[A]_t = [A]_0 - Kt$

$t = 10min \Rightarrow$

$[A]_0 = [A]_0$
$[A]_{10} = [A]_0 - 10K$
$[A]_{20} = [A]_0 - 20K$
$[A]_{30} = [A]_0 - 30K$
⋮

All terms in A.P

(decreasing AP with common diff of $-10K$)

NOTE →

(1) $A \xrightarrow{K}$ Product

$$\frac{-d[A]_t}{dt} = K$$

$$[A]_t = [A]_0 - Kt$$

$$t_{1/2} = \frac{[A]_0}{2K}$$

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IMP ⇒

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eg → $3A \xrightarrow{K} \text{Product}$

Rate const. of rxn

Rate of rxn = $-\frac{1}{3}\frac{d[A]_t}{dt} = K[A]_t^0$

$\Rightarrow -\frac{d[A]_t}{dt} = 3K$

$\Rightarrow [A]_t = [A]_0 - 3K$

$\Rightarrow t_{1/2} = \frac{[A]_0}{2(3K)}$

JEE MAINS 2019 me iske upar ques aaya tha

valid for all order reactions, k replace by mk

★ In general,

$mA \xrightarrow{K} \text{Product}$

Rate const. of rxn

Case won't take this K as rate cont. of A unless it is mentioned.

$[A]_t = [A]_0 - (mK) \times t$

$t_{1/2} = \frac{[A]_0}{2 \times (mK)}$

Special case ⇒

If $mA \xrightarrow{K} \text{Product}$

Rate const. of A (not rate cont. of rxn)

Rate of disapp. of A ⇒ $-\frac{d[A]_t}{dt} = K[A]_t^0$

$\Rightarrow [A]_t = [A]_0 - Kt$

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(2) $A + 2B \xrightarrow{K}$ Product

$$-\frac{1}{1}\frac{d[A]_t}{dt} = -\frac{1}{2}\frac{d[B]_t}{dt} = K[A]_t^0[B]_t^0$$

$$= K$$

After integrating $\Rightarrow$

$$[A]_t = [A]_0 - Kt \;,\; (t_{1/2})_A = \frac{[A]_0}{2K}$$

$$[B]_t = [B]_0 - 2Kt \;,\; (t_{1/2})_B = \frac{[B_0]_0}{2(2K)}$$

If half lives of A & B are equal $\Rightarrow$

$$(t_{1/2})_A = (t_{1/2})_B$$

$$\Rightarrow \frac{[A]_0}{2K} = \frac{[B]_0}{2+K}$$

$$\Rightarrow [B]_0 = 2[A]_0$$

$A + 2B \xrightarrow{K}$ Product

1 : 2 molar ratio is needed for half lives to be equal.

(*) matlab agar A aur B ki half lives equal karni hai toh, we have to take to take them in 1 : 2 molar ratio.

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eg → $A + 2B \xrightarrow{K}$ Products

t=0	1 mol	2 mol
$t = t_{1/2}$	1-0.5	2-1
	= 0.5	= 1

Hence, dono ki half conc. bachi hai bas which will be consumed in another $t_{1/2}$

In this case, half live of rxn = half life of A = half life of B.

$A + 2B \xrightarrow{K}$ Product

Inwl	5 mol.	
$t = t_{1/2}$	t-0.5	5-1
	= 0.5	= 4 mol

→ In this half life of rxn cannot be defined.

(+) half life of reaction tab define hoti hai jab saare reactants apni half conc. use kar chuke ho ek saath.

It can be defined only when half life of all reactants are equal and that will possible only when conc. of all reactants are taken in ratio of their stoichiometric coefficients.

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"for a reaction involving 2 or more reactants, half life of rxn is defined only when the initial conc. of reactants are taken in their molar stoichometric ratios.

In such case,

Half life of rxn = Half life of all reactants "

⇓

This concept is applicable for every order kinetics

Ques: $Cl_2O_4(g)$ decomposes by zero order kinetics as

$$Cl_2O_7(g) \longrightarrow 2ClO_2(g) + 1.5\,O_2(g)$$

Rxn is started w/ initial pressure of $Cl_2O_7(g)$ at 800 Torr & rate of disappearence Cl_2O_7 is 50 Torr/s. Find:

i) Half life of decomposition of Cl_2O_7

ii) Rate of formation of O_2 gas

iii) Total pressure after 10s.

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Soln :

i) $Cl_2O_7(g) \longrightarrow 2ClO_2(g) + 1.5O_2(g)$

Half life : $t_{1/2} = \frac{400}{50} \Rightarrow 8sec$

ii) $\frac{-1}{1}\frac{d[Cl_2O_7]}{dt} = \frac{1}{1.5}\frac{d[O_2]}{dt}$

$\Rightarrow \frac{d[O_2]}{dt} = \frac{3}{2} \times 50\ 25 = 75\ torr/s$

iii) $Cl_2O_7 \rightarrow 2ClO_2 + 1.5O_2$

$t = 0 \quad 800\ Torr \quad 0 \quad 0$

$t = 10s \Rightarrow 800 - 500 \quad 1000 \quad 750$

$\Rightarrow 300\ Torr \quad 1000\ torr \quad 750\ torr$

Total pressure after 10s

$= P(Cl_2O_7) + P(ClO_2) + P(O_2)$

$= 300 + 1000 + 750$

$= 2050\ Torr$

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First order Kinetics ⟹

$$A \xrightarrow{K} \text{Product}$$

$$\boxed{-\frac{d[A]_t}{dt} = K[A]_t^1}$$

↳ Differential Rate law

$$\Rightarrow \int_{[A]_0}^{[A]_t} \frac{d[A]_t}{[A]_t} = -\int_0^t K\,dt$$

$$\Rightarrow \left[\ln[A]_t\right]_{[A]_0}^{[A]_t} = -Kt$$

$$\Rightarrow \ln\left(\frac{[A]_t}{[A]_0}\right) = -Kt$$

$$\Rightarrow \boxed{[A]_t = [A]_0\, e^{-Kt}}$$

↳ Integrated rate law

$$K = \frac{1}{t}\ln\left(\frac{[A]_0}{[A]_t}\right) = \frac{2.303}{t}\log_{10}\left(\frac{[A]_0}{[A]_t}\right)$$

t ⟶

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Recall

At $t=0$, A conc. $= a$

At $t=t$, conc. of A $= a-x$

So, $$K = \frac{2.303}{k} \log_{10}\left(\frac{a}{a-x}\right)$$

Half life for 1st order →

$$t = \frac{1}{K} \ln\left(\frac{[A]_0}{[A]_t}\right)$$

when $t = t_{1/2} \Rightarrow [A]_t = \frac{[A]_0}{2}$

$$\Rightarrow t_{1/2} = \frac{1}{K} \ln\left(\frac{[A]_0}{([A]_0/2)}\right)$$

$$\Rightarrow \boxed{t_{1/2} = \frac{1}{K} \ln(2)}$$

(*) Half life of first order doesn't depend upon on Initial conc.

$$t_{1/2} = \frac{\ln(2)}{K} = \frac{0.693}{K}$$

(*) Half life depends upon temp. as K increases when temp increases, so half life decreases.

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(*) Time of Completion of 1st order
= ∞

(matlab it doesn't complete in a definite time period)

Characteristics of 1st Order Kinetics ⇒

(1) Since rate of reaction is continuously decreasing, so conc. of reactant A reacted in equal time period go on decreasing

Imp (2) Equal fraction or equal ~~fraction~~ percentage reacts in equal time period.

let initial conc. of A = 1 mol/lit,
$t_{1/2}$ = 10 min

100% → 10 min → 50% → 10 min → 25% → 10 min → ⋮

1 mol/lit of A → 10 min → 0.5 mol/lit of A → 10 min → 0.25 mol/lit of A → 10 min → 0.125 mol/lit of A

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(3) conc. of reactant remaining after equal time period constitute a geometric progression.

$$[A]_t = [A]_0 e^{-kt}$$

$$[A]_0 = [A]_0$$

$$[A]_{10} = [A] e^{-10k}$$

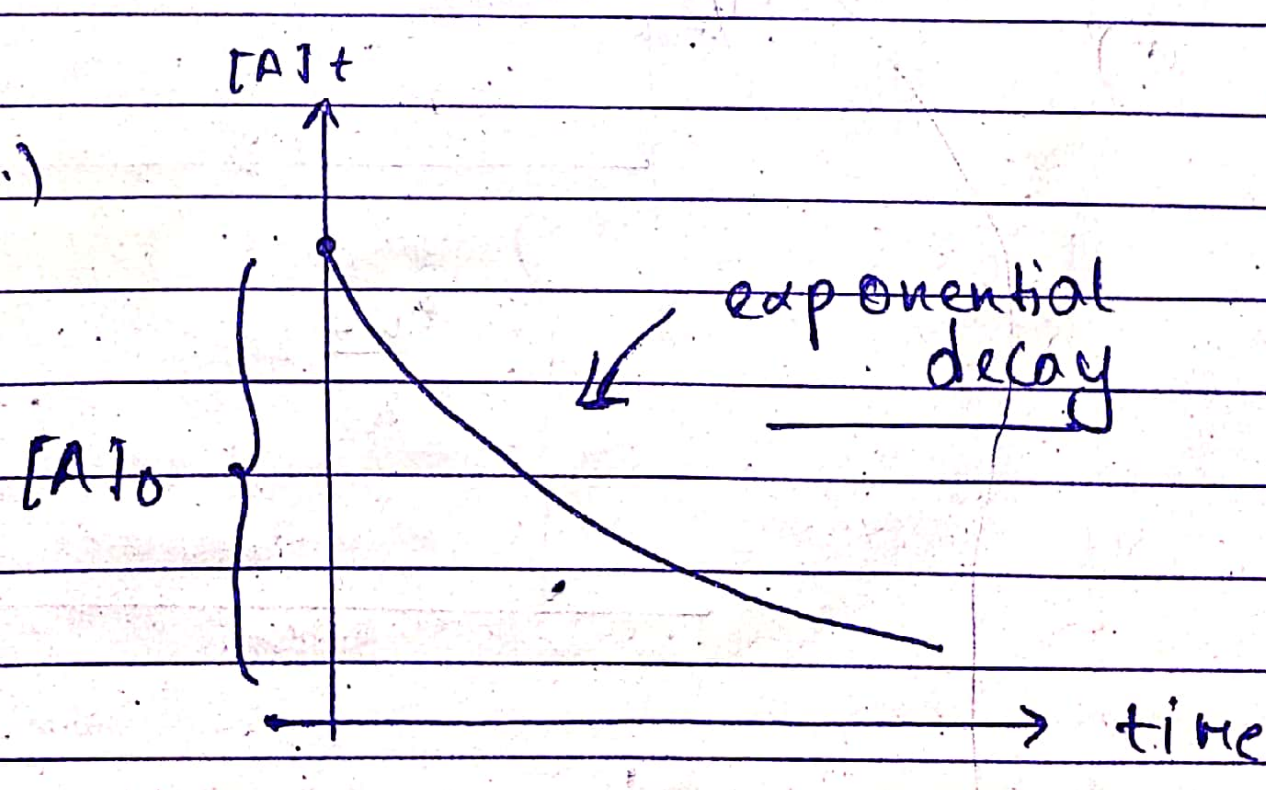
$$[A]_{20} = [A] e^{-20k}$$

⋮

Hence, $[A]_0$, $[A]_{10}$, $[A]_{20}$ ... all terms are in G.P.

(4) Graphs of 1st Order →

a.)



$$\| [A]_t = [A]_0 e^{-kt} \|$$

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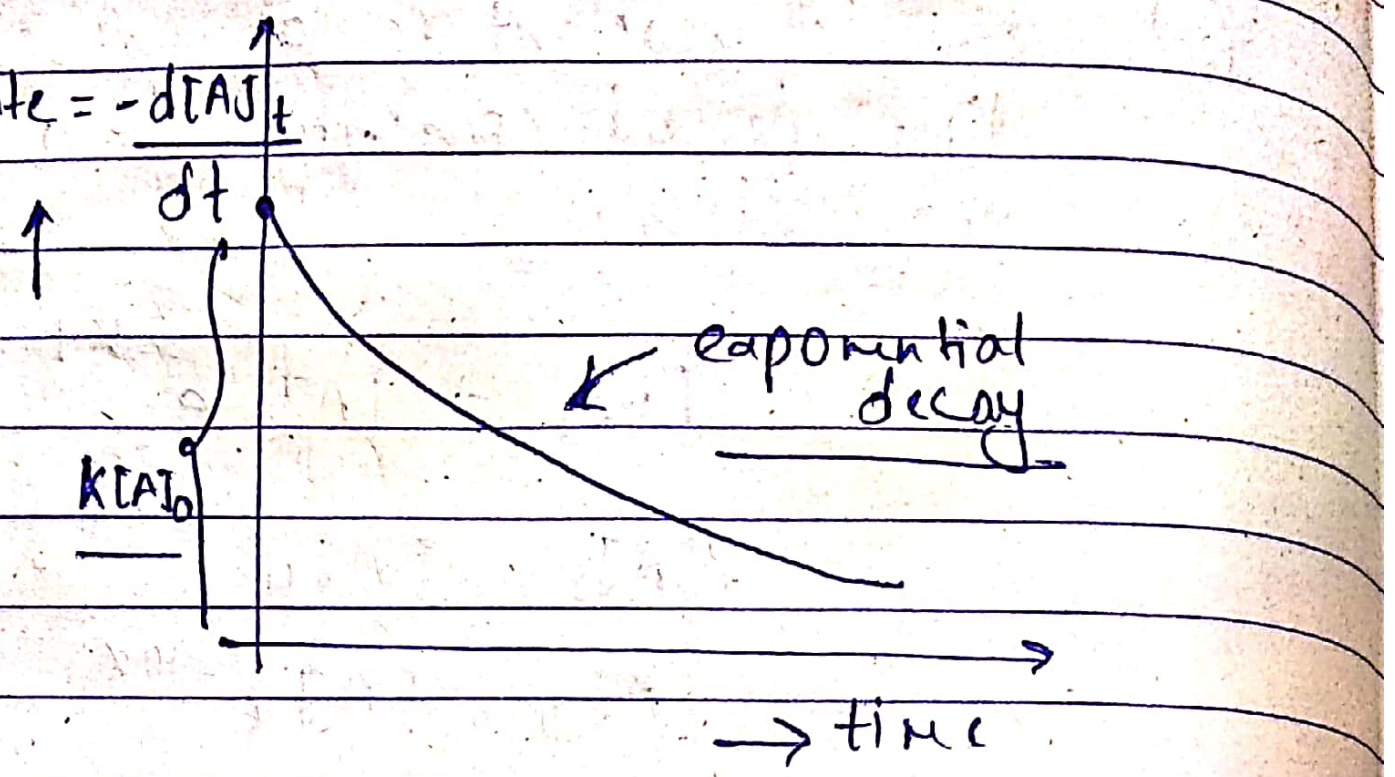
b.) Rate $= -\frac{d[A]_t}{dt}$

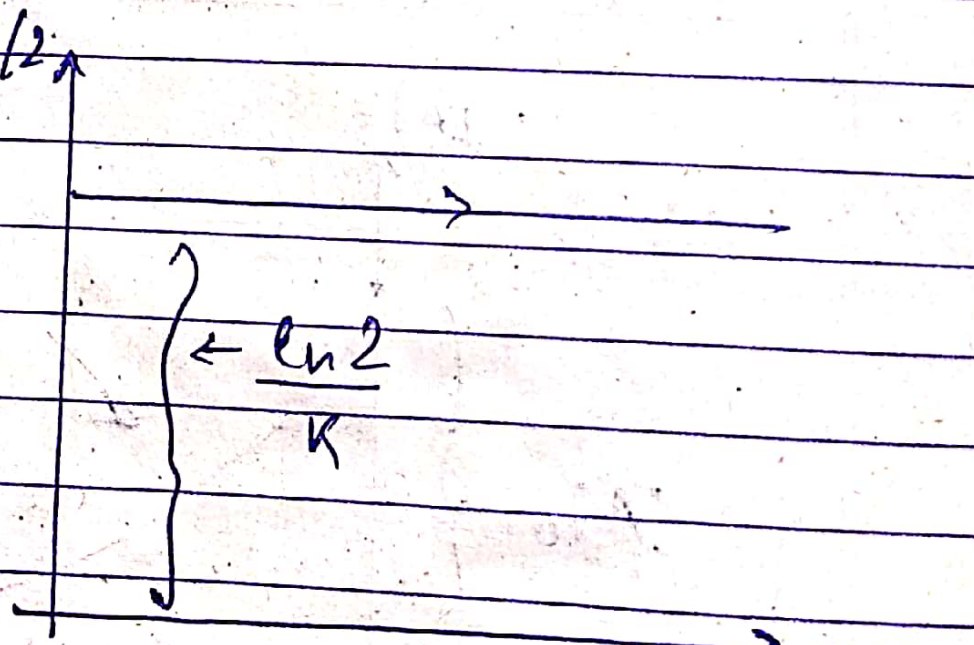


$$[A]_t = [A]_0 e^{-kt}$$

$$\Rightarrow \frac{-d[A]_t}{dt} = [A]_0 (+K) e^{-Kt}$$

$$\Rightarrow \boxed{\frac{-d[A]_t}{dt} = [A]_0 K e^{-kt}}$$

c.)



$$\boxed{t_{1/2} = \frac{\ln 2}{K}}$$

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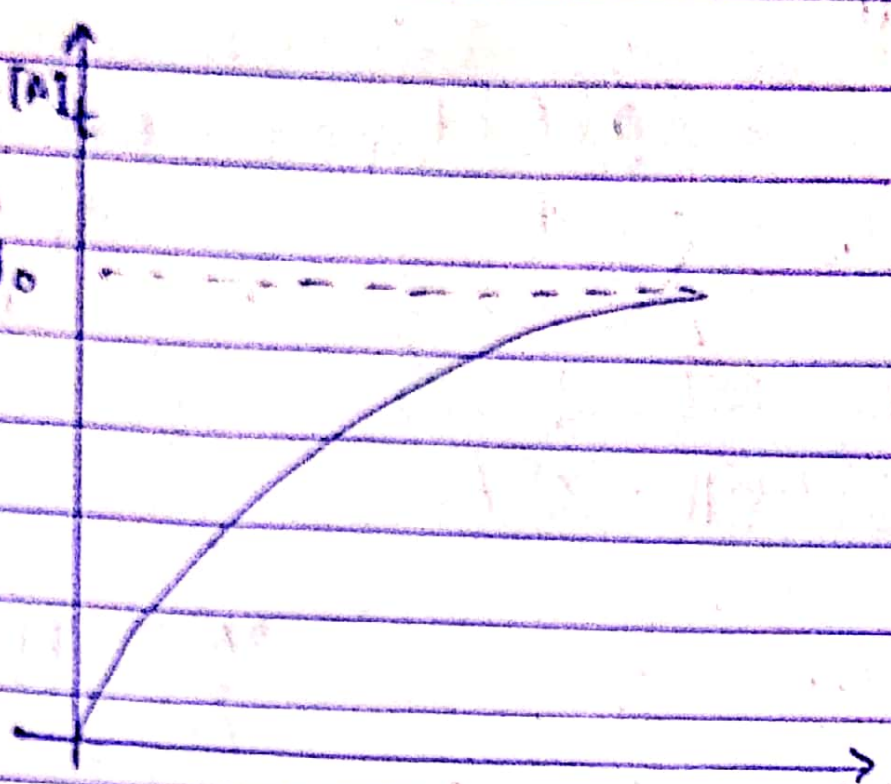
d).



$$x = [A]_0 - [A]_t$$

$$\| x = [A]_0 \left(1 - e^{-kt}\right) \|$$

1st Order Growth Reaction (eg → Bacterial Growth)

$\Rightarrow$

eg → $A \xrightarrow{K} 2A$ { 1st Order Kinetics }

$t = 0$, conc of $A = a$ 0

$t = t$, $a - x$ $2x$

$[A]_0 = a$

$[A]_t = a - x + 2x$

$= (a + x)$

$[A]_t > [A]_0$ } GROWTH

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Here,

$\frac{d[A]_t}{dt}$ = Rate of appearence of A − Rate of disapp. of A

net rate of formation of A

2A is appearing & rate const. = 2K

$$\frac{d[A]_t}{dt} = \underline{2K[A]_t} - \underline{1K[A]_t}$$

↓ A is dec. & its rate const = K

$$\Rightarrow \frac{d[A]_t}{dt} = K[A]_t$$

$$\Rightarrow \int_{[A]_0}^{[A]_t} \frac{d[A]_t}{[A]_t} = k \int_0^t dt$$

$$\Rightarrow \ln\left(\frac{[A]_t}{[A]_0}\right) = Kt$$

$$\Rightarrow K = \frac{1}{t} \ln\left(\frac{[A]_t}{[A_0]}\right)$$

$$\Rightarrow \boxed{K = \frac{1}{t} \ln\left(\frac{a+x}{a}\right)}$$

First Order Growth Eq/n

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NOTE

(1) Initial conc. of reactant = a

For 1st order,
conc. of reactant remaining
after x half lives
$\Rightarrow \left(\frac{a}{2^x}\right)$

(2) • $t_{1/2}$ → half life i.e. half of the reactant has reacted.

Similarly ⇒

• $t_{1/4}$ → $\frac{1}{4}$ th reacted,
$\frac{3}{4}$ th of reactant is remaining

• $t_{1/10}$ → $\frac{1}{10}$ th reacted,
$\frac{9}{10}$ th is remaining

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Ques: For a first order rxn, A → Product, find →

(1) $t_{1/2}$ / $t_{3/4}$ (2) $t_{7/8}$ / $t_{1/4}$

Sol^n

(1) $t_{1/2} = \frac{\ln 2}{k}$

$t_{3/4} = \frac{1}{k} \ln\left(\frac{[A]_0}{([A]_0/4}\right)$

$= \frac{1}{k} \ln(4)$

$\frac{t_{1/2}}{t_{3/4}} = \frac{\ln(2) \times k}{k \quad 2\ln(2)} \Rightarrow \frac{1}{2}$

(2) $\frac{t_{7/8}}{t_{1/4}} = \frac{\frac{1}{k} \ln\left(\frac{[A]_0}{([A]_0/8}\right)}{\frac{1}{k} \ln\left(\frac{[A]_0}{3[A]/4}\right)}$

$\Rightarrow \frac{\ln(8)}{\ln(4/3)}$

$\Rightarrow \frac{3\log(2) \times 2.303}{2.303(2\log 2 - \log 3)}$

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$$\Rightarrow \frac{3 \times 0.3}{2 \times 0.3 - 0.48}$$

$$\Rightarrow \frac{0.9}{0.6 - 0.48}$$

$$\Rightarrow \frac{0.9}{0.12}$$

$$\Rightarrow 7.5$$

Ques: Half life of a 1st order rxn is 10 Mins. Find time in which 99.99% of the reactant is reacted.

Soln →

$$t_{1/2} = 10 = \frac{\ln(2)}{K}$$

$$\| K = \frac{\ln(2)}{10} \|$$

& $t = \frac{1}{k} \ln\left(\frac{100}{100-x}\right)$, where

x = % of reactant reacted

$$t = \frac{10}{\ln(2)} \ln\left(\frac{100}{100-99.99}\right)$$

$$= \frac{10}{\log(2) \times 2.303} \log(10000)$$

$$\Rightarrow \frac{10 \times 3}{0.3} = 100 \text{ Min}$$

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2nd Order Kinetics ⟹

CASE I : $A \xrightarrow{K}$ Products ,

$$-\frac{d[A]_t}{dt} = K[A]_t^2$$

$$\Rightarrow \int_{[A]_0}^{[A]_t} \frac{-d[A]_t}{d[A]_t^2} = k \int_0^t dt$$

$$\Rightarrow \left\{ \frac{1}{[A]_t} \right\}_{[A]_0}^{[A]_t} = kt$$

$$\Rightarrow \boxed{\frac{1}{[A]_t} = \frac{1}{[A]_0} + Kt}$$

↳ Integrated Rate Law

Half life for 2nd order Kinetics ⟹ $(t_{1/2})_2$

when $t = t_{1/2}$, $[A]_t = \frac{[A]_0}{2}$

So, $\frac{1}{[A]_t} = \frac{1}{[A]_0} + Kt$

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$$\Rightarrow \frac{2}{[A]_0} = \frac{1}{[A]_0} + K t_{1/2}$$

$$\Rightarrow \boxed{t_{1/2} = \frac{1}{K[A]_0}}$$

$$t_{1/2} \propto \frac{1}{[A]_0}$$ for 2nd order Kinetics

(+) In general, for n-th order Kinetics $\Rightarrow$

$$(t_{1/2})_n \propto \frac{1}{a^{n-1}}$$

like $\rightarrow$

0-order $(t_{1/2})_0 \propto \frac{1}{a^{-1}} = a$

1st-order $(t_{1/2})_1 \propto \frac{1}{a^0} = 1$ (independent)

2nd-order $(t_{1/2})_2 \propto \frac{1}{a^1}$

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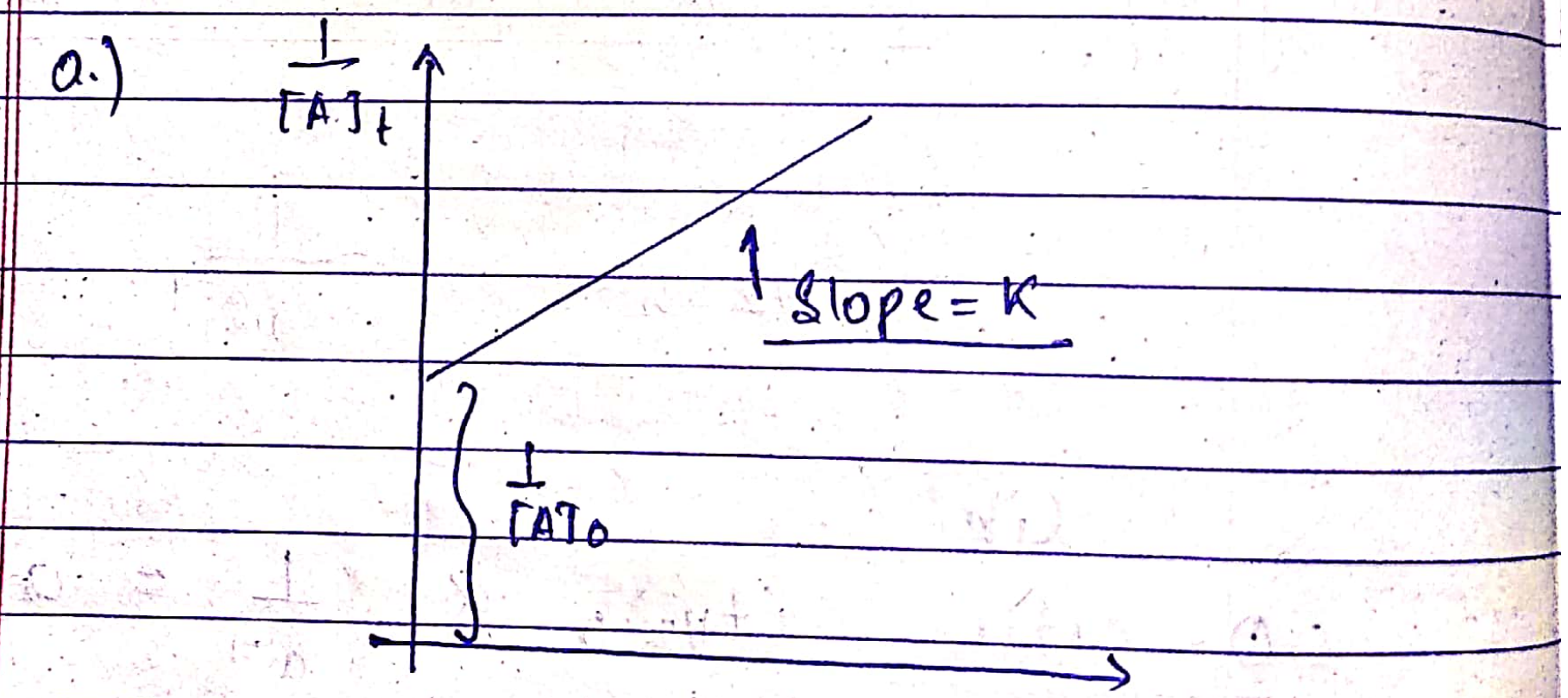
Characteristics of 2nd Order Kinetics →

(A $\xrightarrow{R}$ Products)

(1) It is also not completed in definite time period. (Time of completion $= \infty$)

(2) Graphs of 2nd order →

a.)



$$\frac{1}{[A]_t} = \frac{1}{[A]_0} + Kt$$

$(y = mx + c)$

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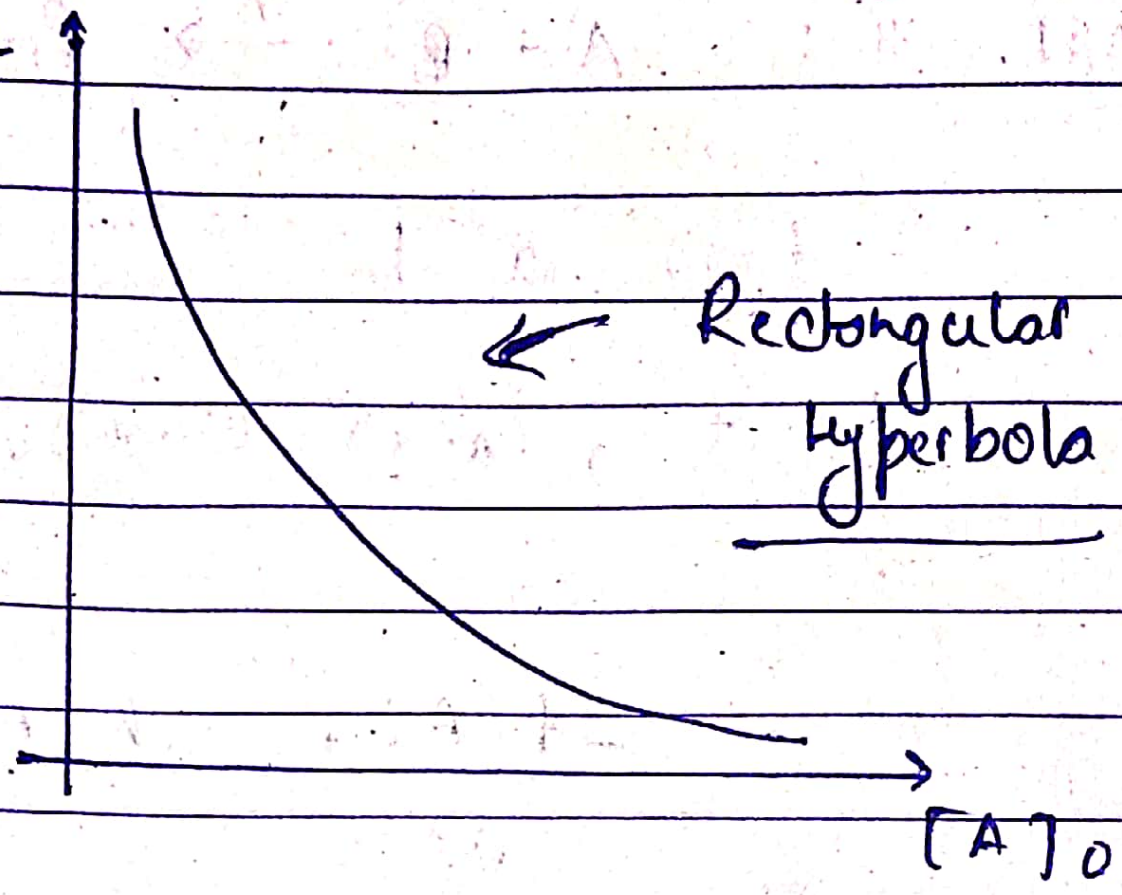
b.)



← Rectangular hyperbola

$$t_{1/2} = \frac{1}{K[A]_0}$$

$$(xy = \text{const})$$

$$\{ [A]_0 \uparrow , (t_{1/2}) \downarrow \}$$

(3) Conc. of reactant remaining after equal time period form an Harmonic progression (H.P.).

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CASE - II : $A + B \xrightarrow{K}$ Products

$t=0 \quad a \quad b$

$t = t \quad (a-x) \quad (b-x)$

$$-\frac{d[A]}{dt}_t = K[A]_t^1[B]_t^1$$

$$\Rightarrow -\frac{d[a-x]}{dt} = K(a-x)(b-x)$$

$$\Rightarrow 0 + \frac{dx}{dt} = K(a-x)(b-x)$$

$$\Rightarrow \int_0^x \frac{dx}{(a-x)(b-x)} = \int_0^t dt \cdot K$$

$$\Rightarrow \frac{1}{(a-b)} \int_0^x \frac{(a-x)-(b-x)}{(a-x)(b-x)} = Kt$$

$$\Rightarrow Kt = \frac{1}{a-b}\left[\int_0^x \frac{1}{b-x}dx - \int_0^x \frac{1}{(a-x}dx\right]$$

$$\Rightarrow t = \frac{1}{(a-b)K}\left[-\ln(b-x) + \ln(a-x)\right]_0^x$$

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$$\Rightarrow t = \frac{1}{(a-b)K}\left[\ln\left(\frac{(a-x)}{(b-x)}\right) - \ln\left(\frac{a}{b}\right)\right]$$

(*) Remember

$$t \Rightarrow \frac{1}{(a-b)K}\left[\ln\left(\frac{b(a-x)}{a(b-x)}\right)\right]$$

NOTE

A + B ⟶ Products

(a) (b)

(a-x) (b-x)

If, $a >> b$, $(a-b) \approx a$

$(a-x) \approx a$

$$\Rightarrow t = \frac{1}{k(a)} \ln\left(\frac{b(a)}{a(b-x)}\right)$$

k'

Such reactions are called Pseudo-order rxns.

First order Kinetics w.r.t B $\Rightarrow$

$$t = \frac{1}{k'} \ln\left(\frac{b}{b-x}\right)$$

↳ Reaction 2nd order se ghat ke 1st order ki ban gayi when we take conc. of A >>> conc. of B

→ Such reactions are called as Pseudo-first order reactions.

Pseudo → false

Now, this 2nd order rxns can be more conveniently studied using 1st order.

eg → A + B → Product

It is found that ⇒

$$Rate = K[A]^1[B]^2$$

If A is constant [as A is taken much larger than B initially]

$$Rate = K'[Y]^2$$

Now, this 3rd order rxn turns into 2nd order & accordingly called as Pseudo-2nd order Rxn

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⟹ When will it be a Pseudo Order Ran? ⟹

⇒ When initial conc. of one of the reactant (1) appearing in rate law is taken much greater than the others.

(2.) If catalyst appears in rate law [Remember conc. of catalyst remains constant during progress of rxn]

(3.) When rate law involves solvent taken in large excess. (because then it will remain constant).

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Examples of Pseudo first order rxn ⇒

① Heterolysis of RCl :

$$RCl_{(aq)} + H_2O_{(l)} \longrightarrow ROH_{(aq)} + HCl_{(aq)}$$

↑ Solvent (excess)

experimentally, Rate = $K[RCl][H_2O]$

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conc. of water = 55.55 M
(can be treated as constant)

$$\Rightarrow \text{Rate} = \underbrace{K[H_2O]}_{\Downarrow K'}[RCl]$$

$$\Rightarrow \text{Rate} = K'[RCl]$$

↑ Pseudo first order rate constant

(2) Hydrolysis of ester in dilute mineral acid ⇒

$$CH_3COOC_2H_5(aq) + H_2O(l) \xrightarrow{H^+} CH_3COOH(aq) + C_2H_5OH(aq.)$$

Experimentally,

$$\text{Rate} = \underbrace{K}_{\text{Const}}[CH_3COOC_2H_5]\underbrace{[H_2O]}_{\text{const as } [H^+]}$$

$$\text{Rate} = K'[CH_3COOC_2H_5]$$

↑ Pseudo first order rate const
$(K' = K[H_2O])$

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(3) Acid catalysed hydrolysis of Sucrose ⇒

$$C_{12}H_{22}O_{11}(aq) + H_2O(l) \xrightarrow[\text{catalyst}]{H^+} \underbrace{C_6H_{12}O_6}_{\text{glucose (d-)}} + \underbrace{C_6H_{12}O_6}_{\text{fructose (l-)}}$$

Both show optical isomerism

$$\text{Rate} = K[C_{12}H_{22}O_{11}][H_2O][H^+]$$

K → rate const. ; $[H_2O]$ → solvent ; $[H^+]$ → catalyst

Here, $K' = K[H_2O][H^+]$

So, $\text{Rate} = K'[C_{12}H_{22}O_{11}]$ —— 1

Pseudo 1st Order Rate Const.

(4) In general, n^{th} order Kinetics ⇒ —— II

$$A \xrightarrow{K} \text{Products}$$

$$-\frac{d[A]_t}{dt} = K[A]_t^n$$

$$\Rightarrow -\int_{[A]_0}^{[A]_t} \frac{d[A]_t}{[A]_t^n} = K\int_0^t dt$$

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$$\Rightarrow -\left[\frac{[A]_t^{-n+1}}{-n+1}\right]_{[A]_0}^{[A]_t} = kt \qquad \text{when } n!=1$$

$$\Rightarrow \frac{1}{(n-1)}\left[\frac{1}{[A]_t^{n-1}} - \frac{1}{[A]_0^{n-1}}\right] = kt$$

$$\Rightarrow \boxed{t = \frac{1}{k(n-1)}\left[\frac{1}{[A]_t^{n-1}} - \frac{1}{[A]_0^{n-1}}\right]}$$

|| n^{th} order kinetics eqⁿ ||

Applicable (except n=1)

Half life ⇒

when $t = t_{1/2}$, $[A]_t = \frac{[A]_0}{2}$

$$(t_{1/2})_n = \frac{1}{k(n-1)}\left[\frac{1 \times 2^{n-1}}{[A]_0^{n-1}} - \frac{1}{[A]_0^{n-1}}\right]$$

$$\| (t_{1/2})_n = \frac{1}{k(n-1)} \frac{(2^{n-1}-1)}{[A]_0^{n-1}} \|$$

as ⇒ $(t_{1/2})_n \propto \frac{1}{[A]_0^{n-1}}$

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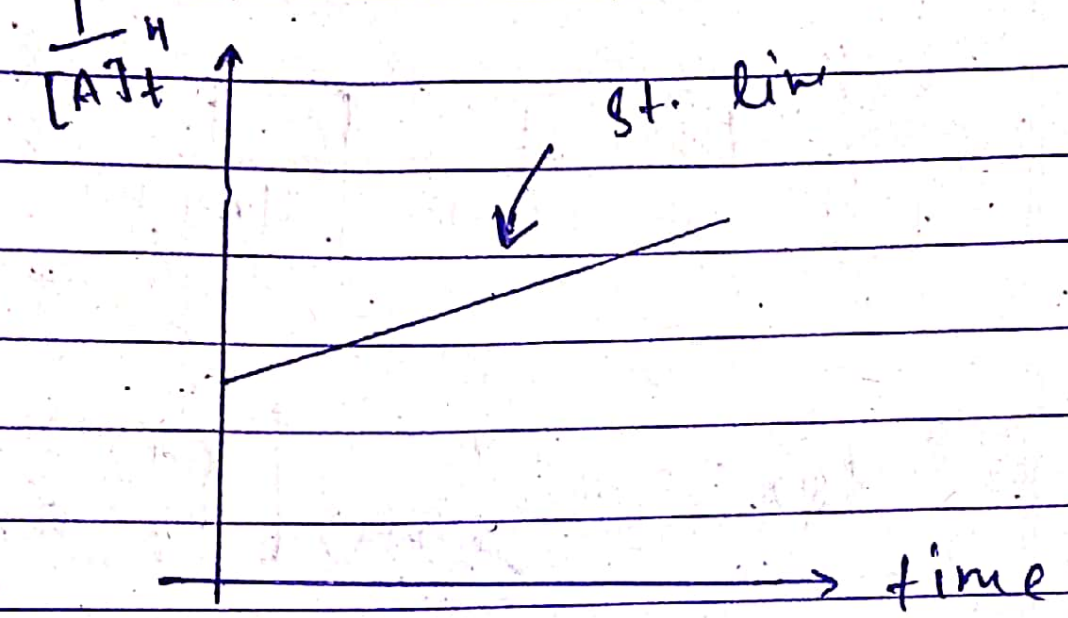
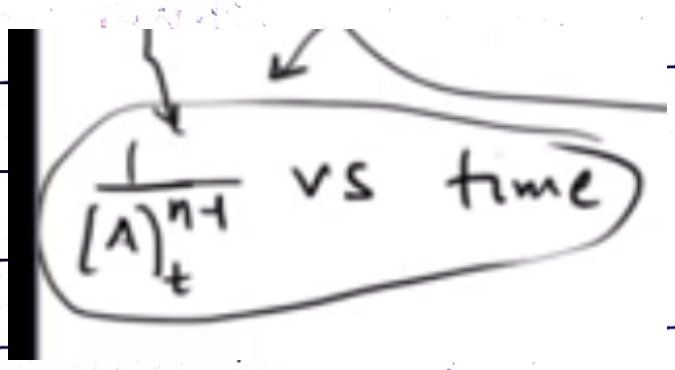
Ques For a rxn, A ⟶ Product, variation in conc. of A with time is found as ⇒



Find order of rxn.

Soln

graph of  will be a straight line for nth order reaction

$n - 1 = 4$

$n = 5$

∴ 5th order Kinetics

(*) So, $\frac{1}{[A]_t^{n-1}}$ vs. time ka graph humesha ek st. line banega.

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Ques Find $\frac{t_{3/4}}{t_{1/2}}$ for n^{th} order rxn

$A \longrightarrow$ Products

$$t = \frac{1}{K(n-1)} \left[\frac{1}{[A]_t^{n-1}} - \frac{1}{[A]_0^{n-1}} \right]$$

$$\frac{t_{3/4}}{t_{1/2}} = \frac{\frac{1}{K(n-1)} \left[\frac{(4)^{n-1}}{[A]_0^{n-1}} - \frac{1}{[A]_0^{n-1}} \right]}{\frac{1}{K(n-1)} \frac{(2^{n-1}-1)}{[A]_0^{n-1}}}$$

$$\Rightarrow \frac{4^{n-1} - 1}{2^{n-1} - 1}$$

$$\Rightarrow \frac{2^{2(n-1)} - 1}{2^{n-1} - 1}$$

$$\Rightarrow \frac{(2^{n-1}+1)(2^{n-1}-1)}{(2^{n-1}-1)}$$

$$\Rightarrow 2^{n-1} + 1$$

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Ques find conc. of reactant after 30 mins if 20% of reactant is disassociated in first 10 mins of initial conc. Rxn is 1st order. Initial conc. of reactant = 10 M.

initial conc. is 10 M

Sol^n In first order, equal fraction/ percentage reacts in equal time period.

First 10 Mins → 80% of 10 M remains

= 8 M

Conc. of reactant after 30 Min

⇒ 8 × 0.8 × 0.8

⇒ 5.12 M

remaining

Pehle 10 Min me → 80%.

Dusre 10 Min me → bache hue ka 80%.

Teesre 10 Min me → uska bhi 80%.

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Ques Oxidation of a certain metal is found to obey the equation →

$$\tau^2 = \alpha t + \beta$$

where τ is the thickness of oxide film, at time t. α, β are constant. Find order of rxn.

Soln

$$2\tau \cdot \frac{d\tau}{dt} = \alpha + 0$$

$$\Rightarrow \frac{d\tau}{dt} = \frac{\alpha (\tau)^{-1}}{2}$$

order = -1

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Experimental verification Of Order Of Reaction ⇒

(1) Initial Rate method :

(V. IMP for JEE Mains, Advanced & Boards)

For every n reactants, n+1 experiments are done.

$A + B \longrightarrow$ Products

$$\text{Rate} = K[A]^p[B]^q \quad -(1)$$

Expt. No.	$[A]_0$	$[B]_0$	initial rate
1	2×10^{-2} M	10^{-3} M	5×10^{-3} M Min
2	2×10^{-2} M	4×10^{-3} M	1×10^{-2} M Mi
3	4×10^{-2} M	10^{-3} M	2×10^{-2} M Min

Using above experimental values, we can find partial values of p & q

Using (1) & (2) $\Rightarrow q$ $p = \frac{1}{2}$

(1) & (3) $\Rightarrow$ $p = 2$

Order of reaction $= \frac{5}{2}$

=

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(2) Half life method :

In general, $(t_{1/2})_{n^{th}} \propto \frac{1}{a^{n-1}}$

where, a = initial conc. of rxn
n = order of rxn

$$\frac{(t_{1/2})_2}{(t_{1/2})_1} = \left(\frac{a_1}{a_2}\right)^{n-1}$$

$$\Rightarrow (n-1)\{\log a_1 - \log a_2\} = \log(t_{1/2})_2 - \log(t_{1/2})_1$$

$$\Rightarrow n = 1 + \frac{\log(t_{1/2})_2 - \log(t_{1/2})_1}{\log a_1 - \log a_2}$$

eg Q: $[A]_0 = 1\ M$, $t_{1/2} = 10\ Min$

when $\Rightarrow 2M$, $(t_{1/2}) = 10 Min$

Soln. Since, half life is constant not depending on initial conc.,
hence, 1st order rxn

eg:

$[A]_0 \longrightarrow 10^{-3}\,M$, $(t_{1/2}) = 10$ Min

$\downarrow$

$10^{-4}\,M$, $(t_{1/2}) = 100$ Min

$\longrightarrow$ 2nd order rxn

Note

Order of rxn determined from half life method can be applied to the rxn types like ⇒

(i) $A \longrightarrow$ Product

(ii) $A + 2B \longrightarrow$ Product

(1 : 2 → initial conc. should be in 1:2 molar ratio

Quesn

$$\underset{(g)}{2NO} + \underset{(g)}{2H_2} \longrightarrow \underset{(g)}{N_2} + \underset{(g)}{2H_2O}$$

When equimolar mixture of NO & H_2 is taken at initial total pressure of 350 torr, total pressure reduces to half its initial value in ~~how many~~ minutes. 121 minutes. When initial total pressure is taken 275 torr, half life is 196 min.

Find order of rxn.

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Sol^{n} $P_1 = 350$ torr, $t_{1/2} = 121$ min

$P_2 = 275$ torr, $t_{1/2} = 196$ min

$$\Rightarrow \frac{196}{121} = \left(\frac{350}{275}\right)^{n-1}$$

$$\Rightarrow \frac{196}{121} = \left(\frac{14}{11}\right)^{n-1}$$

$$\Rightarrow 2 = n-1$$

$$\Rightarrow n = 3$$

$\therefore$ • 3rd order rxn

(3) Integrated Rate law method $\Rightarrow$
(Hit & Trial Method)

eg. A $\longrightarrow$ Products.

Data given,
Time $= t_1, t_2, t_3, t_4 \ldots$
$[A]_t = a_1, a_2, a_3, a_4 \ldots$

Find order.

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Assume, this data fits in zero order.

$$[A]_t = [A]_0 - kt$$

$$\Rightarrow K = \frac{[A]_0 - [A]_t}{t - 0}$$

$$K_1 = \frac{a_1 - a_2}{t_2 - t_1} \quad \Big| \quad K_2 = \frac{a_2 - a_3}{t_3 - t_2} \quad \ldots\ldots$$

→ If K_1 is nearly equal to K_2, then we can say that data fits well in zero order integrated law, hence rxn is zero order.

→ If $K_1 \neq K_2$, calculated above, then try your luck with first order.

$$K = \frac{2.303}{t} \log \left(\frac{[A]_0}{[A]_t} \right)$$

$$K_1 = \frac{2.303}{t_2 - t_1} \ln \frac{a_1}{a_2} \quad \Big| \quad K_2 = \frac{2.303}{t_3 - t_1} \ln \left(\frac{a_1}{a_3} \right)$$

- - - .

If $K_1 \cong K_2$, we can say

reaction is first order.

Report value of K as → $K = \frac{K_1 + K_2 + \ldots K_n}{(\text{avg.})^n}$

report the average value of all the values of k obtained

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Monitoring the progress of first order rxn ⇒

(Determination of rate const. = K)

(1) By measurement of pressure (applicable for gaseous rxn)

(2) By Titration (for soln phase rxn's) (or volumetric analysis)

(3) By measurement of optical rotatio

(1) By Pressure Measurement →

→ Applicable for rxn (1st order involving gases (with / without solid liquid)

→ At const. vol. & temperature

$P \propto n$ [$PV = nRT$] for gases

or $P \propto$ conc. (C)

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→ Measurement of pressure of gas is much easier than the determination of conc. of gas.

eg $\underset{(g)}{A} \longrightarrow \underset{(g)}{B} + \underset{(g)}{2C}$

↑ Rxn is 1st order wrt. A.

$t=0$	a	0	0
$t=t$	$a-x$	x	$2x$

here, x is the moles of A consumed

$$K = \frac{1}{t} \ln\left(\frac{a}{a-x}\right)$$

[a → initial conc. of A]

⇒ Progress of this 1st order rxn is monitored by measuring total pressure.

Data collected,

$t=0$; t

Total pressure = P_0 , P_t find K

$A \longrightarrow B + 2C$

$t=0$,	P_0	0	0
$t=t$	P_0-x	x	$2x$

here, x is the pressure of gas reacted

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Total pressure = P_T

$$\Rightarrow P_T = P_0 - x + x + 2x$$

$$\Rightarrow x = \frac{(P_t - P_0)}{2}$$

$$K = \frac{1}{t} \ln\left(\frac{P_0}{P_0 - x}\right)$$

$$K = \frac{1}{t} \ln\left(\frac{P_0}{P_0 - \frac{(P_t - P_0)}{2}}\right)$$

$$K = \frac{1}{t} \ln\left(\frac{2P_0}{3P_0 - P_t}\right)$$

General expression,

$$A_n(g) \longrightarrow nA(g)$$

[1st order wrt A_n]

Initially, an inert gas & $A_n(g)$ are taken in a vessel of fixed vol.

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Data recorded :

$t = 0 \;,\; t \;,\; \infty$

Total pressure $= P_0 \;,\; P_t \;,\; P_\infty$
(P_t)

Find K.

Sol^n.

$$A_n(g) \xrightarrow{K} nA(g)$$

	A_n (g)	nA (g)	Total pressure
$t=0$	p	0	$p + p_{He} = p_0$
$t=t$	$p-x$	nx	$p-x+nx = p_t$
$t=\infty$	$p-p$	np	$p-p+np = p_\infty$

$$p + p_{He} = p_0 \quad \text{—— (1)}$$

$$p - x + nx + p_{He} = p_t \quad \text{—— (2)}$$

$$0 + np + p_{He} = p_\infty \quad \text{—— (3)}$$

$$K = \frac{1}{t} \ln\left(\frac{p}{p-x}\right)$$

(3) - (1) $\Rightarrow$

$$p(n-1) = p_\infty - p_0$$

(3) - (2) $\Rightarrow$

$$(n-1)(p-x) = p_\infty - p_0$$

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Divide (3)-(2) & (3)-(1)

$$\frac{P(n-1)}{(n-1)(p-x)} = \frac{P\infty - P_0}{P\infty - P_t}$$

$$\Rightarrow \frac{p}{p-x} = \frac{P\infty - P_0}{P\infty - P_t}$$

Remember so, $K = \frac{1}{t} \ln\left(\frac{P\infty - P_0}{P\infty - P_t}\right)$

⇓

General formula

Ques^n

NH_4NO_2 soln decomposed as

(aq)

first order decomposition of

NH_4NO_2 as $NH_4NO_2 \xrightarrow{K} N_2 + 2H_2O$

(aq) (g) (l)

This reaction is monitored by Total pressure measurement

time	$t=0$	20 min	∞
Press:	= 50 mm Hg	80 mm	90 mm

$$K = \frac{1}{t}\ln\left(\frac{P\infty - P_0}{P\infty - P_t}\right) = \frac{1}{t}\ln\left(\frac{90-50}{90-80}\right)$$

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$$K = \frac{2\ln(2)}{20}$$

$$\Rightarrow \frac{\ln(2)}{t_{1/2}} = \frac{2\ln(2)}{20}$$

$$\Rightarrow t_{1/2} = 10 \text{ min}$$

(2) **By Titration** : (Generally, in soln phase)

eg.) First order decomposition of H_2O_2 (aq)

$$H_2O_{2(aq)} \xrightarrow{K} H_2O_{(aq)} + \frac{1}{2}O_{2(aq)}$$

[1st order wrt H_2O_2]

$t=0$	a	0	0
$t=t$	$a-x$		

$$K = \frac{1}{t}\ln\left(\frac{a}{a-x}\right)$$ } We have to find a & (a-x)

→ Progress of 1st order decomposition of H_2O_2 can be monitored by titration with $KMnO_4$ solution in acidic medium.

$$KMnO_4 + H^+ + H_2O_2 \longrightarrow O_2 + Mn^{2+}$$

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Data Recorded ⇒

	$t = 0$	,	t
Vol of $KMnO_4$ used	$= V_0$	,	V_t

At $t=0$, no. of equivalents of H_2O_2 = no. of equivalents of $KMnO_4$

$$\therefore V_{H_2O_2} \times 2 = V_{KMnO_4} \times C \times 5$$

$$= V_0$$

At $t=t$, $(a-x) \times V_{H_2O} \times 2 = V_t \times C \times 5$

We can see that →

$$\therefore V_0 \propto (a)$$

$$V_t \propto (a-x)$$

$$\Rightarrow \frac{V_0}{V_t} = \frac{a}{a-x}$$

$$\boxed{K = \frac{1}{t} \ln\left(\frac{V_0}{V_t}\right)}$$

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4MP

eg: **Acid catalysed hydrolysed of ester:**

$$CH_3COOC_2H_5 + H_2O \xrightarrow[K \text{ catalyst}]{H^+} CH_3COOH + C_2H_5OH$$

	$CH_3COOC_2H_5$ (aq)	H_2O (l)		
$t=0$	a			
$t=t$	$a-x$		CH_3COOH (aq) x	C_2H_5OH (aq) x
$t=\infty$	0			

[1st order wrt $CH_3COOC_2H_5$]

Progress of this rxn is monitored by standard NaOH soln ⇒

Data	$t=0$	t	∞
Vol. of NaOH	V_0	V_t	V_∞

$t=0$, $V_0 \propto$ Catalyst concentration

$t=t$, $V_t \propto$ Conc. (catalyst) + x

$t=\infty$, $V_\infty \propto$ C(catalyst) + a

$$\frac{V_\infty - V_0}{V_\infty - V_t} = \frac{a}{a-x}$$

$$K = \frac{1}{t} \ln\left(\frac{a}{a-x}\right) \equiv \frac{1}{t} \ln\left(\frac{V_\infty - V_0}{V_\infty - V_t}\right)$$

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eg:) Iodination of Acetone ⇒

$\underset{\text{Acetone}}{CH_3COCH_3} + I_2 \longrightarrow CH_3COCH_2I + HI$

∴ Reaction is 1st order w.r.t Acetone

$t=0$	a	C_{I_2}	0	0
$t=t$	$(a-x)$	$(C_{I_2}-x)$	x	x
$t=\infty$	0	$(C_{I_2}-a)$	a	a

→ Via equivalent concept, Hypo (standard) is used to monitor progress of rxns involving I_2.

$CH_3COCH_3 + I_2 \longrightarrow CH_3COCH_2I + HI$

$t=0$	a	C_{I_2}	0	0
$t=t$	$(a-x)$	$(C_{I_2}-x)$	x	x
$t=\infty$	0	$(C_{I_2}-a)$	a	a

Data: $t = 0, t, \infty$

Vol. of Hypo V_0, V_t, V_∞

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|| # STUDY ['EQUIVALENT CONCEPT'] TO UNDERSTAND THIS ||

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: Hypo will react only with iodine, hence,

$$V_0 \propto C_{I_2}$$

$$V_t \propto (C_{I_2} - x)$$

$$V_\infty \propto (C_{I_2} - a)$$

$$K = \frac{1}{t} \ln\left(\frac{a}{a-x}\right)$$

$$V_0 - V_\infty \propto a$$

$$V_0 > V_t > V_\infty$$

$$V_t - V_\infty \propto (a-x)$$

$$|| \quad K = \frac{1}{t} \ln\left(\frac{V_0 - V_\infty}{V_t - V_\infty}\right) \quad ||$$

Q1. Decomposition of H_2O_2 is a first order rxn. A soln of H_2O_2 ~~is~~ ~~levelled~~ labelled as '20 vol' was left open. Due to this some H_2O_2 decomposed. To determine the new volume strength after 6hrs, 10ml of this soln was diluted to 100ml. 10 ml of this diluted soln was titrated against 25ml of 0.025N $KMnO_4$ soln under acidic conditions. Cal. rate const. of decomposition of H_2O_2.

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Sol^n

$$H_2O_2\ (aq) \xrightarrow{K} H_2O(\ell) + \frac{1}{2}O_2\ (g)$$

$t=0$ a

$t=6$ hrs $(a-x) = [A]_t$ $a = [H_2O_2]_0 = \frac{20}{11.2}$ M

$\frac{V_t = 2}{}$

$KMnO_4 + H^+ + H_2O_2\ (aq) \rightarrow O_2 + Mn^{4}$

Equivalents of $KMnO_4$ = Equivalents of H_2O_2

$\Rightarrow 25 \times 0.025 \times 5 = 0.1 \times [A]_t \times 10 \times 2$

$\Rightarrow [A]_t = \frac{125 \times 0.025}{2}$

$K = \frac{1}{6} \ln\left(\frac{20/11.2}{\frac{125 \times 0.025}{2}}\right)$ hr^{-1}

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Imp for Advance

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Q→ A certain reactant B^{n+} is getting converted to $B^{(n+4)+}$ in soln by 1st order reaction. Rate const. of this rxn is measured by titrating a vol. of this soln with reducing agent which only reacts with B^{n+} & $B^{(n+4)+}$. In this process, it converts B^{n+} to $B^{(n-2)+}$ & $B^{(n+4)+}$ to $B^{(n-1)+}$. At t=0, the vol. of reagent consumed is 25ml and t=10min, the vol. used up is 32ml. Find rate const (K) of conversion of B^{n+} to $B^{(n+4)+}$.

Soln.

$$B^{n+}_{(aq)} \xrightarrow{K} B^{(n+4)+}_{(aq)}$$

	B^{n+}	$B^{(n+4)+}$
t=0	a	0
t=10min	a-x	x

$B^{n+} \xrightarrow{R.A.} B^{(n-2)+}$ v.f. = 2

$B^{(n+4)+} + RA \longrightarrow B^{(n)+}$ v.f. = 5

$25 = a \times 2 + 0 \Rightarrow$

$a = 12.5$

$32 = (a-x) \times 2 + x \times 5$

$32 = 25 + 3x$

$x = 7/3$

$$K = \frac{1}{10} \times 2.303 \times \log\left(\frac{12.5}{12.5 - 7/3}\right) = 0.2303 \times \log\left(\frac{45}{61}\right)$$

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(3) **By measurement of optical rotation:**

(for rxn involving optically reactive compound(s))

eg.) Inversion of Cane sugar (i.e. Sucrose) (or hydrolysis of sugar) ($C_{12}H_{22}O_{11}$)

(+)

$$C_{12}H_{22}O_{11}(aq) + H_2O(l) \xrightarrow{k} C_6H_{12}O_6 + C_6H_{12}O_6$$

(d) (d) (+) (l) (+)

(+) indicates that compound is optically active.

[1st Order in Sucrose]

→ Optical rotation by an optically active compound, $\theta \propto$ conc. of that substance x thickness of polarimeter

Observed Rotation

$\theta = \alpha c t$

$\theta = r \times c \times t$ ← length of polarimeter tube

r → specific rotation of cmpd. (doesn't depend upon conc. of optically active compounds)

c → conc. (in mol/L)

Take t = unity (1)

$\Rightarrow \theta = r \times c$

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General expression :

$$A^{*} \xrightarrow{K} B^{*} + C^{*} \qquad \text{Rate} = K[A]^{1}$$

$t=0$	a	0	0
$t=t$	$a-x$	x	x
$t=\infty$	0	a	a

Date Recorded ⇒ 1

time : 0, t, ∞

Total rotation : θ_0, θ_t, θ_∞

→ Assuming r_A, r_B, r_C are specific rotation of A, B, C respectively.

$$\theta_0 = a \times r_A$$

$$\theta_t = (a-x) \times r_A + x \times r_B + x \times r_C$$

$$\theta_\infty = a \times r_B + a \times r_C$$

$$\frac{\theta_\infty - \theta_0}{\theta_\infty - \theta_t} = \frac{a(r_B + r_C - r_A)}{(a-x)(r_B + r_C - r_A)}$$

$$K = \frac{1}{t} \ln\left(\frac{a}{a-x}\right) \Rightarrow \frac{a}{a-x} = \frac{\theta_\infty - \theta_0}{\theta_\infty - \theta_t}$$

$$\Rightarrow K = \frac{1}{t} \ln\left(\frac{\theta_\infty - \theta_0}{\theta_\infty - \theta_t}\right)$$ Remember

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$$K = \frac{1}{t} \ln\left(\frac{\theta_\infty - \theta_0}{\theta_\infty - \theta_t}\right)$$

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Above formula is applicable for 1st order rxn like ⇒

$$\rightarrow A^{*} \longrightarrow B^{*} + C^{*}$$

$$\rightarrow A^{*} \xrightarrow{D^{*}} B + 2C$$

↳ D is optically active catalyst.

Q. $A \longrightarrow B + C$, first order wrt A. All A, B & C are optically active. Observed rotation of the mixture after 20mins was 5° while after completion of reactions, it was −20°. If optical rotation per mole of A, B, C are 60°, 40° & −80° respectively, find half-life & average life of this 1st order rxn.

Soln:

	A^{*}	$\longrightarrow$	B^{*}	$+ C^{*}$	$r_A = 60°$
					$r_B = 40°$
					$r_C = -80°$
$t = 0$	a		0	0	
$t = 20$ Mins	$(a-x)$		x	x	
$t = t_\infty$	0		a	a	

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At $t = 20$ mins $\Rightarrow$

$$S = (a-x)r_A + x \times r_B + x \times r_C$$
$$= (a-x)60 + 40 \times x + x \times (-80)$$

At $t = t_\infty \Rightarrow$

$$-20 = 0 \times 60 + a \times 40 + a \times (-80)$$

$$\Rightarrow a = \frac{1}{2}$$

In $t = 20$ mins $\Rightarrow$

$$S = 60a - 80x$$

$$80x$$

$$S = 60a - 100x$$

$$100x = 30 - S$$

$$\boxed{x = \frac{1}{4}}$$

$$K = \frac{\ln(2)}{t_{1/2}} = \frac{1}{20}\frac{\ln(1/2)}{\left(\frac{1}{2} - \frac{1}{4}\right)}$$

$$\boxed{t_{1/2} = 20 \text{ min}}$$

$$(T_{avg})_{1^{st}\ order} = \frac{1}{K} = \frac{1}{\frac{(\ln 2)}{(K_{1/2})}} \Rightarrow \frac{20}{0.693}$$

$$\Rightarrow 28.8 \text{ Min}$$

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NOTE: Calculation of avg. life or Mean life of rxn (T_{avg}) →

$$T_{avg} = \frac{\text{Total life of all molecules of reactant}}{\text{Total no. of reactant molecules}}$$

$$= \frac{(-dN_1) \times t_1 + (-dN_2) \times t_2 + (-dN_3) \times t_3 + \dots}{N_0}$$

$$= \frac{\int_0^{t_{completion}} (-dN \times t)}{N_0 \text{ (no. of molecules)}}$$

$(-dN)$ ⇒ no. of molecules of reactant which lived for t time & then decayed

(*)

For first Order

$$T_{avg} = \frac{\int_0^{\infty} (-d[A]_0) \times t}{[A]_0}$$

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$$\left.\begin{aligned} [A]_t &= [A]_0 e^{-kt} \\ \Rightarrow \quad -d[A]_t &= k[A]_0 e^{-kt} dt \end{aligned}\right\}$$

$$\Rightarrow \quad \tau_{avg} = \frac{\int_0^\infty \{k[A]_0 e^{-kt} dt\} \times t}{[A]_0}$$

$$\Rightarrow \quad \tau_{avg} = k \int_0^\infty t e^{-kt} dt$$

$$= k \times \frac{1}{k^2} = \frac{1}{k}$$

$$=$$

Avg. life for 1st order (τ_{avg})

$$= \frac{1}{k}$$

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(2) Effect of temperature on Rate of Rxn ⟹

→ For most chemical rxn, rate of reaction become double to triple for marly every 10° rise in temperat

Collision Theory ⟹

1.) for a rxn to proceed, reactants must collide with each other

2.) Every collision isn't fruitful or successful collision. For a collision to be successful, reactant molecules must possess a certain minimum total energy called Threshold energy and also the proper orientation at the time of collision.

3.) Reactants are believed to be converted to products via the formation of a hypothetical species, called Transition State (T.S.) or Activated complex, in which bonds in reactant are not completely broken & the bonds in products are not completely formed.

T.S. cannot be experimentally

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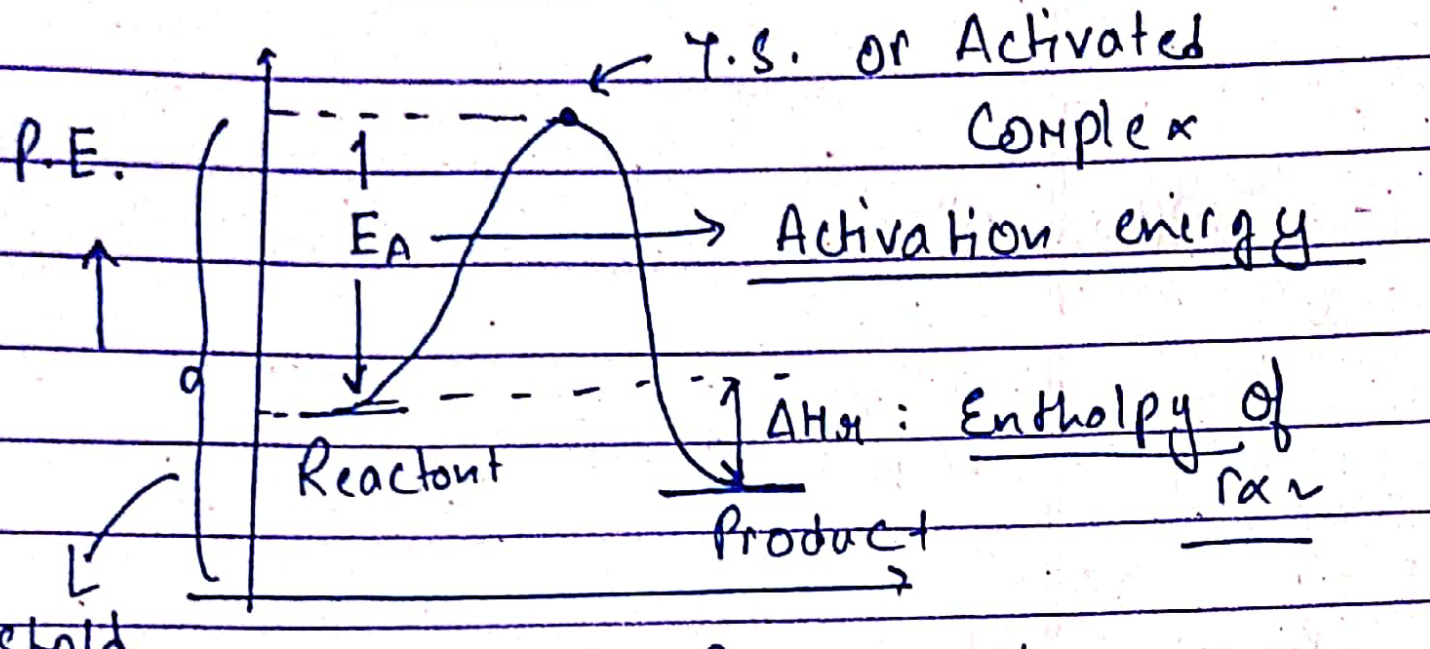
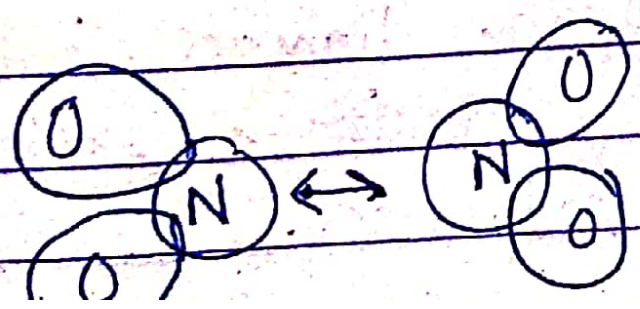
studied as it can't be isolated (due to very high energy).



(*) Threshold energy → The minimum total energy of reactant molecules required to cross energy barrier to form product species (provided collision occurs with proper orientation).

(*) Activation Energy (E_A) → The additional K.E. to be possessed by reactant molecules above the avg. energy of reactants to cross the energy barrier with proper orientation.

eg-) $NO_2 + NO_2 \longrightarrow N_2O_4$ DIMER



N_2O_4 } PROPER COLLISION

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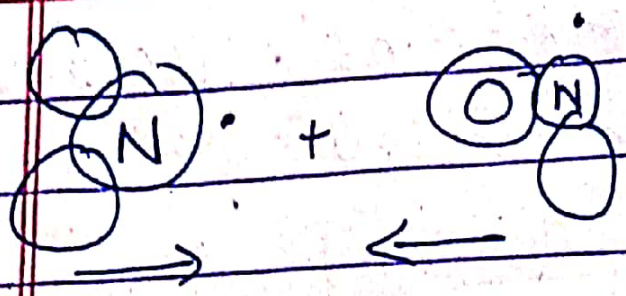
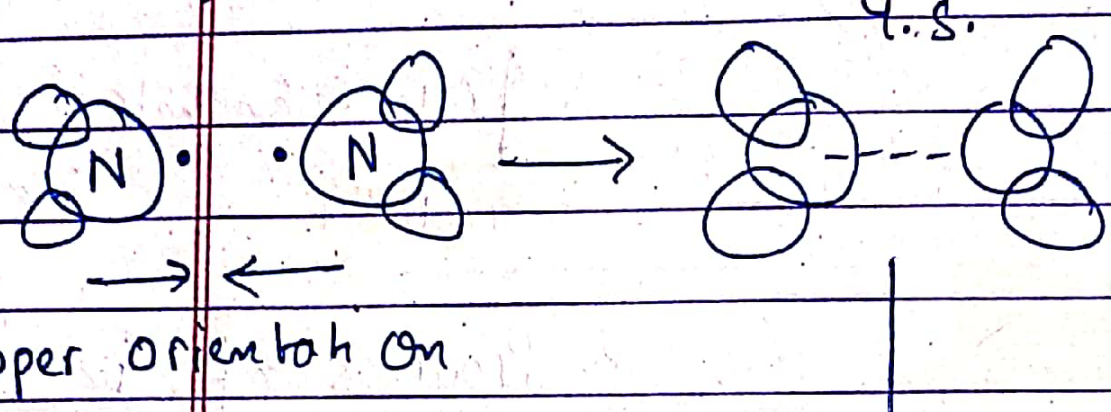
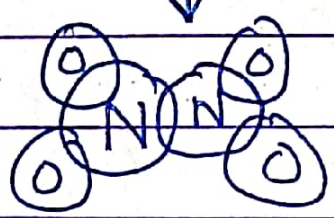


IMPROPER COLLISION : No Product

Proper Collision ⇒



Proper orientation



Product (N_2O_4)

effectively,

✓ Rate of reaction = Collision frequency × Energy barrier factor × Orientation factor (or steric factor) (or Probability factor)

- Collision frequency (Z_{AB}) (i.e. no. of collisions per unit time per unit volum

$$= \pi \sigma_{AB}^2 \times \sqrt{\frac{8kT}{\pi \mu_{AB}}} \times \left(\frac{N_A}{V}\right) \times \left(\frac{N_B}{V}\right)$$

Where, σ_{AB} = Collision diameter

$k = R / N_A$ = Boltzman const.

μ_{AB} = Reduced mass = $\frac{M_A \times M_B}{M_A + M_B}$

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No. of
N_A = molecules of A
N_B = " B
V = Volume of vessel

- Energy Barrier factor ~~factor~~ takes care of fraction of reactant molecules which have energy greater than or equal to or equal to E_A.

OR

fraction of molecules which have energy $\geq E_A$, $\propto \frac{e^{-E_A/RT}}{\sqrt{T}}$ {T → temp in Kelvin}

Or, approximately, fraction of molecules having energy $\geq E_a = e^{-E_A/RT}$

- Orientation factor : usually less than 1, but can be equal to 1 for atomic combinations.

(*) → Collision frequency is a large value whilst, energy barrier factor & orientation factor are very small values.

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$$\Rightarrow Z_{AB} = \pi \sigma_{AB}^2 \times \sqrt{\frac{8kT}{\mu_{AB}}} \times \left(\frac{N_A}{V}\right) \times \left(\frac{N_B}{V}\right)$$

$\pi \sigma_{AB}^2$ → const. for a given system.

Conc. terms (Conc. of reactants)

$\sqrt{\frac{8kT}{\mu_{AB}}}$: $\sqrt{T}$ is a very small term.

$\Rightarrow \frac{e^{-E_a/RT}}{\sqrt{T}}$

Hence,

$\Rightarrow$ Rate of reaction $= \left(A e^{-E_a/RT}\right) \times (\text{conc.})^{\text{Order}}$

Constant

& $K = A e^{-E_a/RT}$ — Arrhenius eqn

Rate constant.

Frequency factor of pre-exponential factor

$A = Z_{AB} \times P$

Frequency factor — Collision frequency — Steric factor (or orientation factor)

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Arrhenius Equation →

$$K = A e^{-Ea/RT}$$

Rate const.

& A → frequency factor / pre-exponential factor

Ea → Activation energy of rxn

R → Gas const.

T → temp (in K)

Rate of rxn = K (conc.)$^{\text{order}}$

——→ | —— | ——

Note : (1) Max value of rate const. K = A

When, (i) ⇒ temp ⟶ ∞

(ii) ⇒ Ea = 0

(as in case of free radical combination)

|| $\dot{C}H_3 + \dot{C}H_3 \longrightarrow C_2H_6$ ||

(2) Activation energy, Ea, can be +ve or zero but never -ve.

(3) A & Ea are taken const. (independent of temperature)

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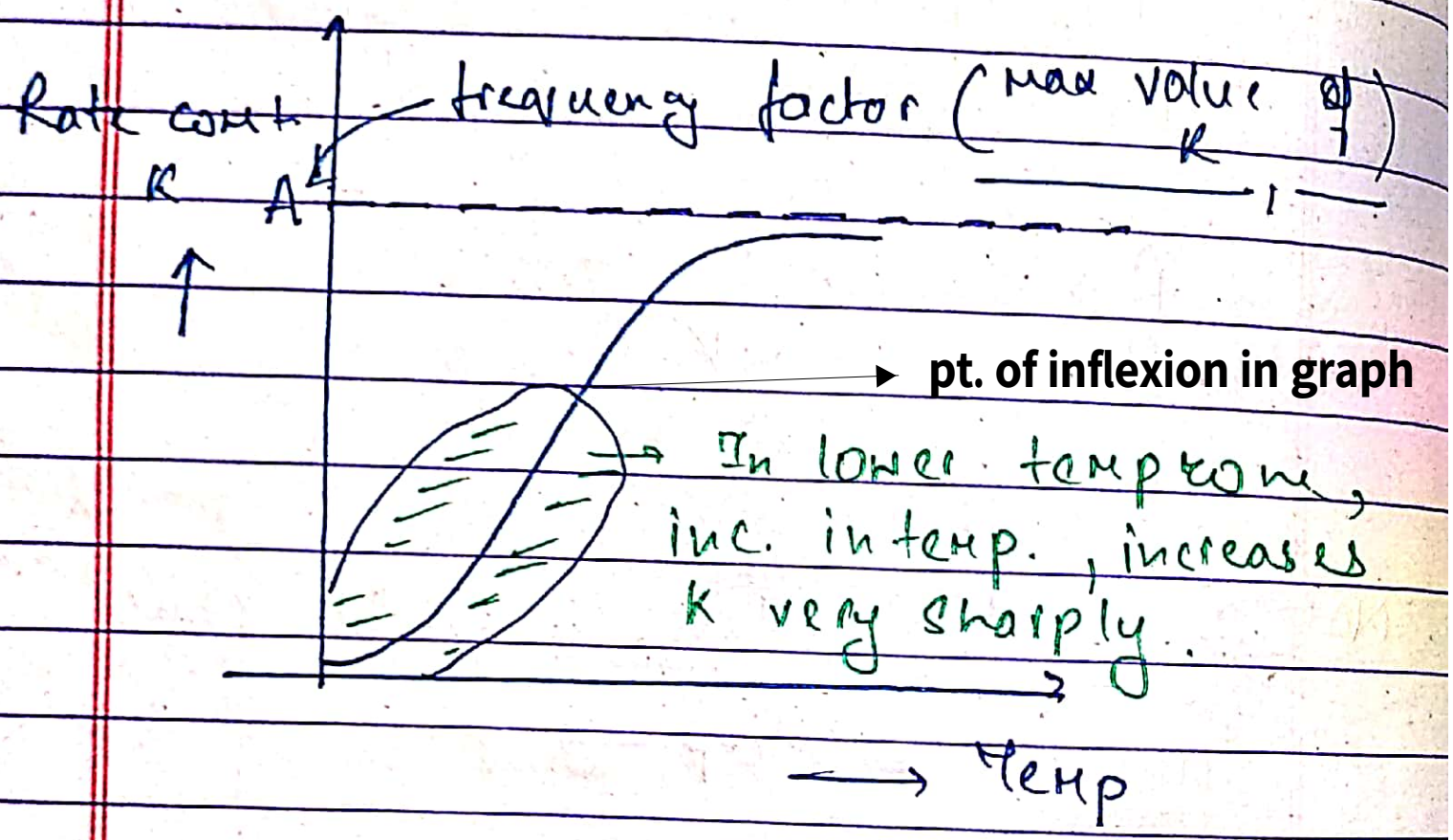
$K = A e^{-Ea/RT}$	$T \uparrow$, $K \uparrow$ (sharply) Rate $\uparrow$



- **Integrated form of Arrhenius Eqn ⇒**

$$K = A e^{-Ea/RT}$$

$$\Rightarrow \ln(K) = \ln(A) - \frac{Ea}{RT}$$

At temp $T_1 \Rightarrow \ln(K_1) = \ln(A) - \frac{Ea}{RT_1}$ — (1)

" $T_2 = \ln(K_2) = \ln(A) - \frac{Ea}{RT_2}$ — (2)

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(2) - (1)

$$\Rightarrow \ln\left(\frac{K_2}{K_1}\right) = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\Rightarrow \log_{10}\left(\frac{K_2}{K_1}\right) = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

Integrated Arrhenius eqⁿ || Remember ||

Q. At 300K, rate const. of a rxn $= 2 \times 10^{-3}$ find Rate const. at 320K if activation energy of rxn is 23.03R kJ/mol.

Sol^n

$$\log \frac{K_2}{(2\times10^{-3})} = \frac{23.03R \times 10^3}{2.303R}\left\{\frac{1}{300} - \frac{1}{320}\right]$$

$$\log\left(\frac{K_2}{2\times10^{-3}}\right) = 10000 \times \frac{20}{300\times32} = \frac{5}{24}$$

Solve it.

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• Differential form of Arrhenius Eqⁿ ⇒

$$K = A e^{-Ea/RT}$$

$$\Rightarrow \ln(K) = \ln(A) - \frac{Ea}{RT}$$

Differentiate wrt T

$$\Rightarrow \frac{1}{K}\frac{d(K)}{dt} = 0 - \frac{Ea}{R}\left(\frac{-1}{T^2}\right)$$

$$\text{Or} \Rightarrow \boxed{d(\ln K) = \frac{Ea\ dt}{RT^2}}$$

↓

Differential form of Arrhenius Eqⁿ

→ Applications of Arrhenius Eqⁿ ⇒

(i.) Consider two rxns →

$$A \xrightarrow{Ea_1} \text{Product} \quad , \quad B \xrightarrow{Ea_2} \text{Product}$$

eg. If $Ea_1 > Ea_2$ ~ faster
, A ↑ slow rex. B $\xrightarrow{Ea_2}$

→ Both rxns occuring at same temp & temp of both rxns is increased by some value,

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i.e., $T \rightarrow$ same, $dT \rightarrow$ same

(eg → 300K → 301K)

$T = 300K$, $dT = 1K$

bono reactions same temp pe chal rahi hai, & dono ka temp amount (dT) se badhaya hai ⇒

$$\frac{d(\ln K)}{} = \frac{Ea}{RT^2} dt$$

$$\Rightarrow d(\ln K) \propto Ea$$

(imp)

We can say that reaction with higher value of Ea is more sensitive towards temperature variation.

OR With rise in temp., rate of constant, K, of that reaction increases more sharply, which has higher value.

(fraction increase / % increase)

V. Imp ⇒

(i) If $Ea_1 > Ea_2$, & frequency factor (A) of both rxn's are same, then Ea_1 wali slow rxn hogi & Ea_2 will faster rxn.

(Choto barrier hai toh → more molecules will cross the barrier → rxn is faster

(ii) Consider a rxn →

$Ea = 100\ KJ/mol$

$X \longrightarrow Y$

→ this react is studied wrt to temp rise at two diff. temp.

(1) 300K ⟶ 301 K
(2) 400K ⟶ 401 K

dT = same
T = different
Ea → same

$$d(\ln K) = \frac{Ea}{RT^2} \; dT$$

Ea → same
dT → same
RT^2 → const.

$$\Rightarrow \boxed{d(\ln K) \propto \frac{1}{T^2}}$$

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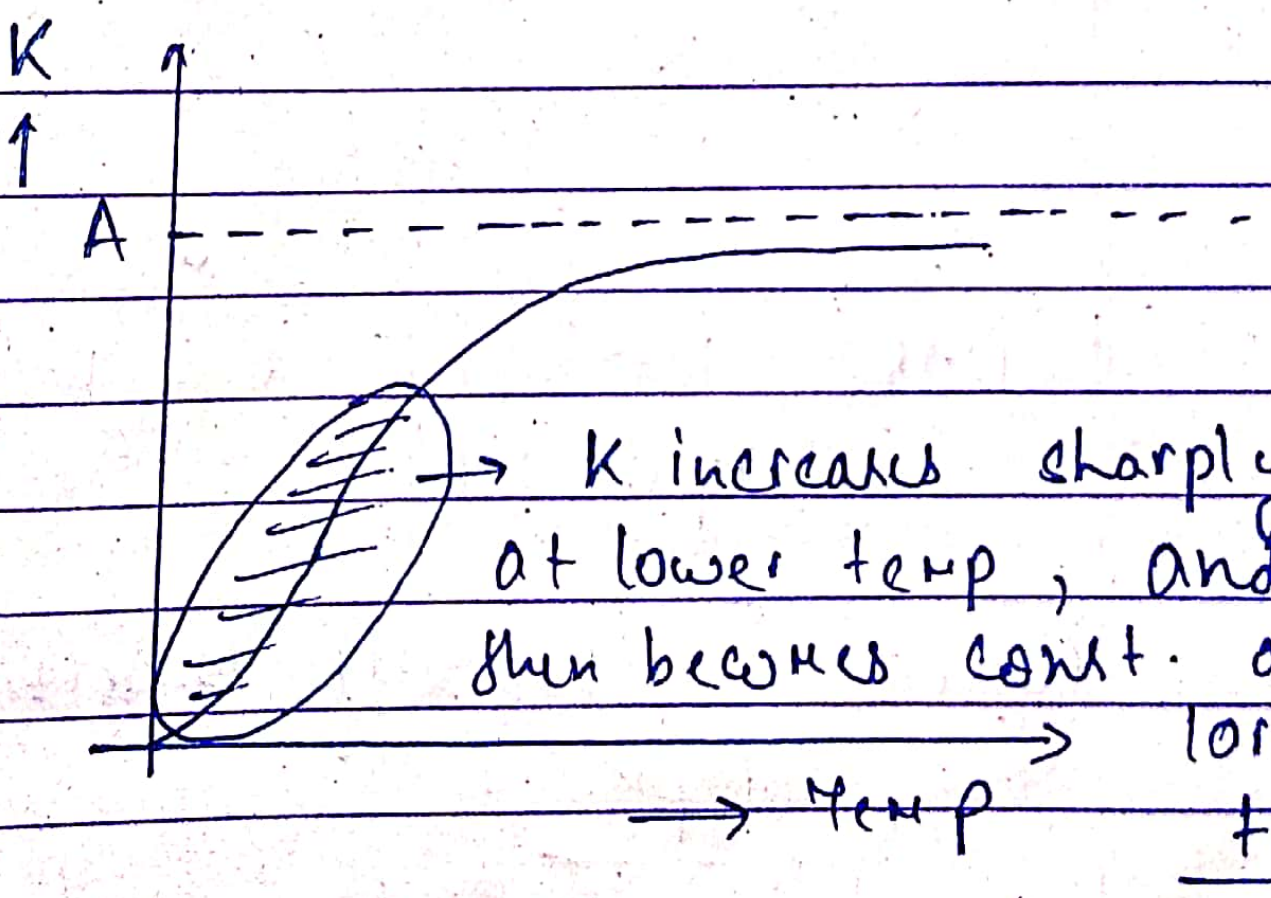
Imp ⟹

→ Rate const. of a rxn increase more sharply with rise in temp at lower temp.

Hence, rate const. will increase more sharply at 300 K ⟶ 301 K rather than 400 K ⟶ 401 K.

He can also see, this by graph →



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→ Q. Why does rate of reaction becomes double to triple for every 10° rise in temp!

Ans

$KE \propto T$

$T \rightarrow T + 10$

$300K \rightarrow 310K$

% change in avg KE ⇒

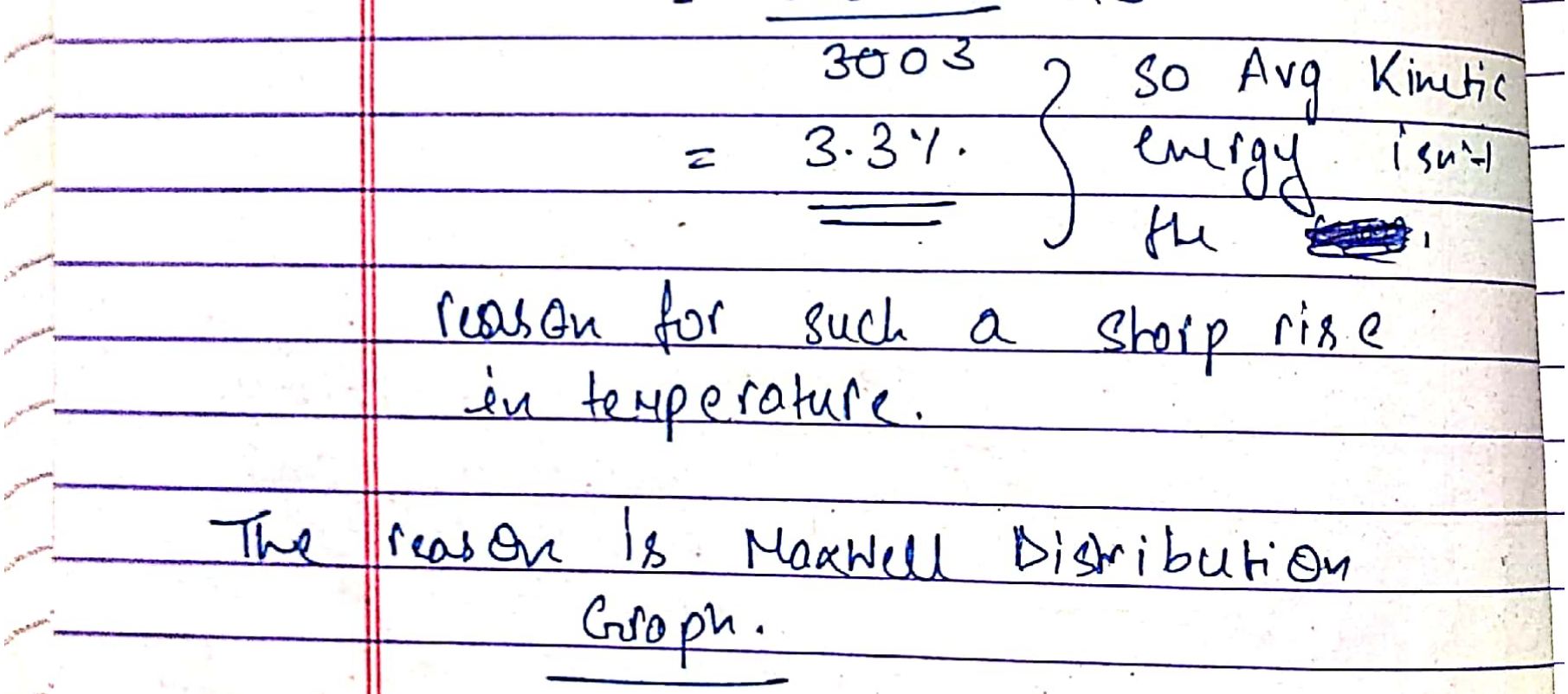
$= \frac{310 - 300}{3003} \times 100$

$= 3.3\%$

So Avg Kinetic energy is not the reason for such a sharp rise in temperature.

The reason is Maxwell Distribution Graph.

Maxwell Distribution of KE →



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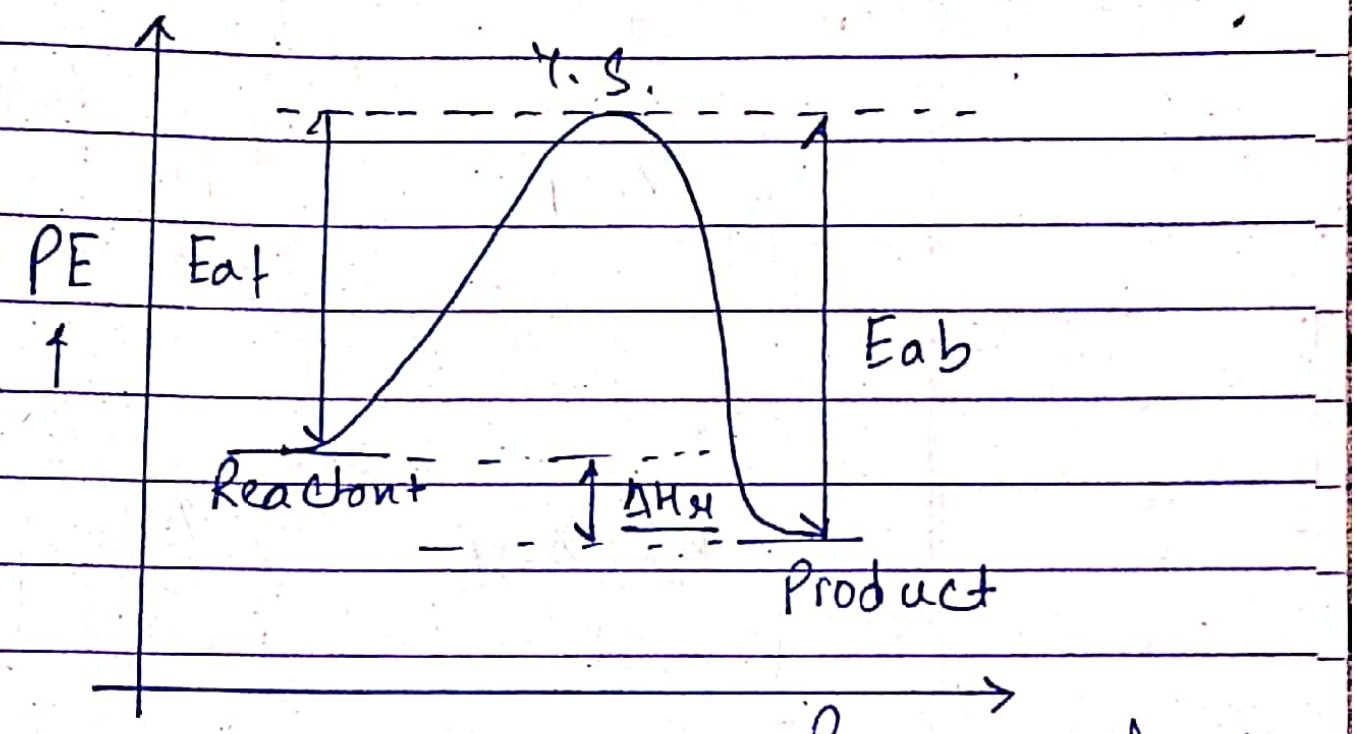
Case of Reversible reaction :

$$A \underset{K_b}{\overset{K_f}{\rightleftharpoons}} B \quad , \quad K_{eq} = \frac{K_f}{K_b}$$

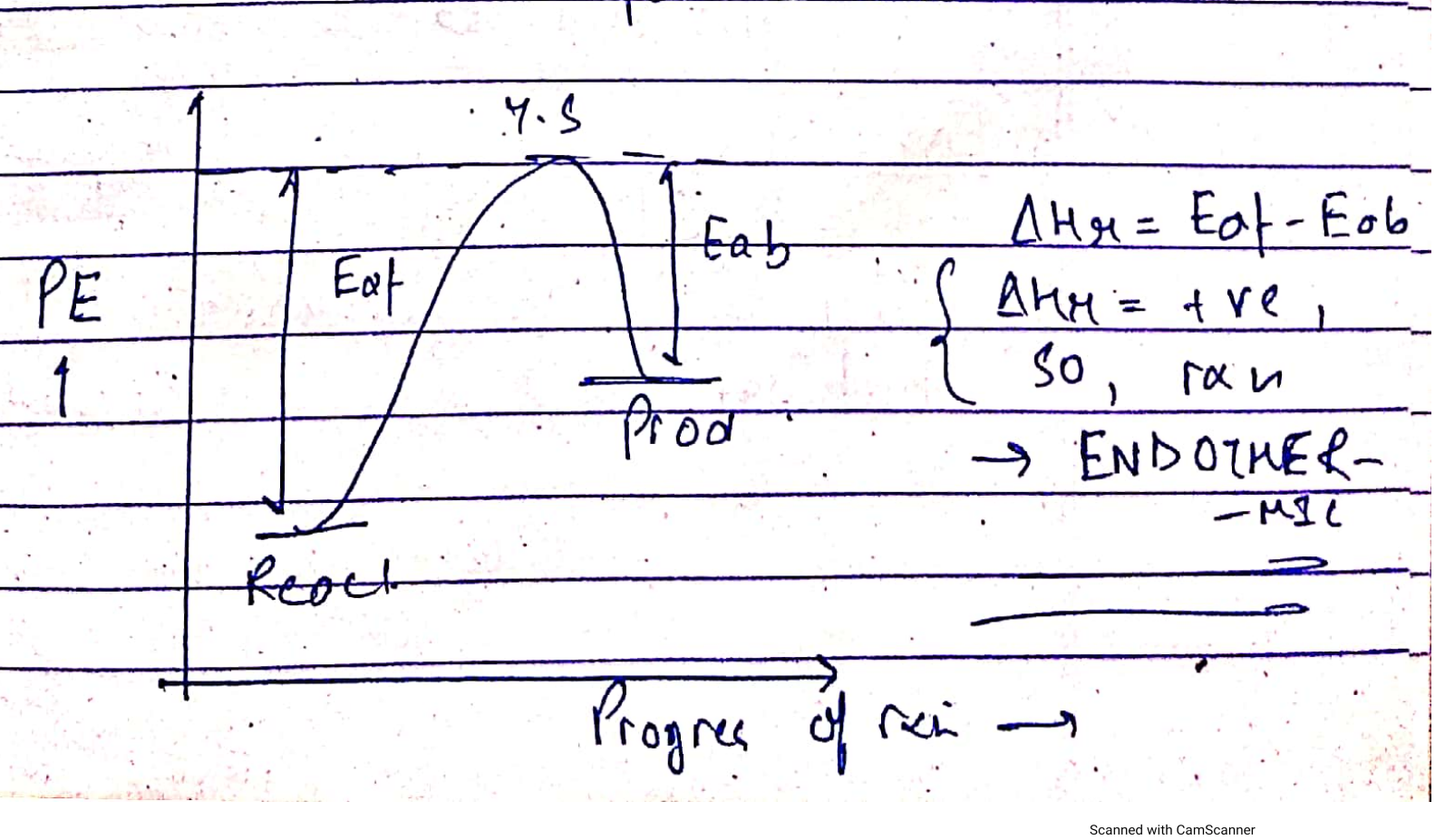


YMP

$$\boxed{\Delta H_R = E_{af} - E_{ab}}$$

ΔH_R = -ve as $E_{ab} > E_{af}$ so, EXOTHERMIC

where, E_{af} → activation energy for forward diri
E_{ab} → activation energy for backward rxn.



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$$K_f = A_f e^{-E_{af}/RT}$$

$$K_b = A_b e^{-E_{ab}/RT}$$

$$\Rightarrow \frac{K_f}{K_b} = \frac{A_f}{A_b} e^{-(E_{af} - E_{ab})/RT}$$

$$\Downarrow$$

$$\Rightarrow \boxed{K_{eq} = \frac{A_f}{A_b} e^{-\Delta H_R/RT}}$$

Van't Hoff Eqn.

$$\ln \frac{(K_{eq})_{T_2}}{(K_{eq})_{T_1}} = \frac{\Delta H_R}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

Ques

$A_{(g)} \rightleftharpoons B_{(g)}$, $\Delta H_R = -33\ kJ/mol$ at 300K

Find (i.) Eqlbm const

(ii.) find E_{af} & E_{ab} if $\frac{E_{af}}{E_{ab}} = \frac{20}{31}$, Take frequency factor of both forward & backward rxn. to be equal

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Solu

(1) $K_{eq} = \frac{K_f}{K_f} = \frac{A_f}{A_b} e^{-\Delta H_R / RT}$ [as $A_f = A_b$]

$= e^{(33 \times 10^3) / 8.314 \times 300}$

$= e^{330/3 \times 8.314}$

$= e$

(2) $\Delta H_R = E_{af} - E_{ab}$

$-33 = E_{af} - \frac{31}{20} \times E_{af}$

$\Rightarrow \frac{-11}{20} E_{af}$

$\Rightarrow E_{af} = 20 \times 3 = 60$ kJ/mol.

$\therefore E_{ab} = 93$ kJ/mol.

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(3) **Effect of Catalyst on rate of reaction ⇒**

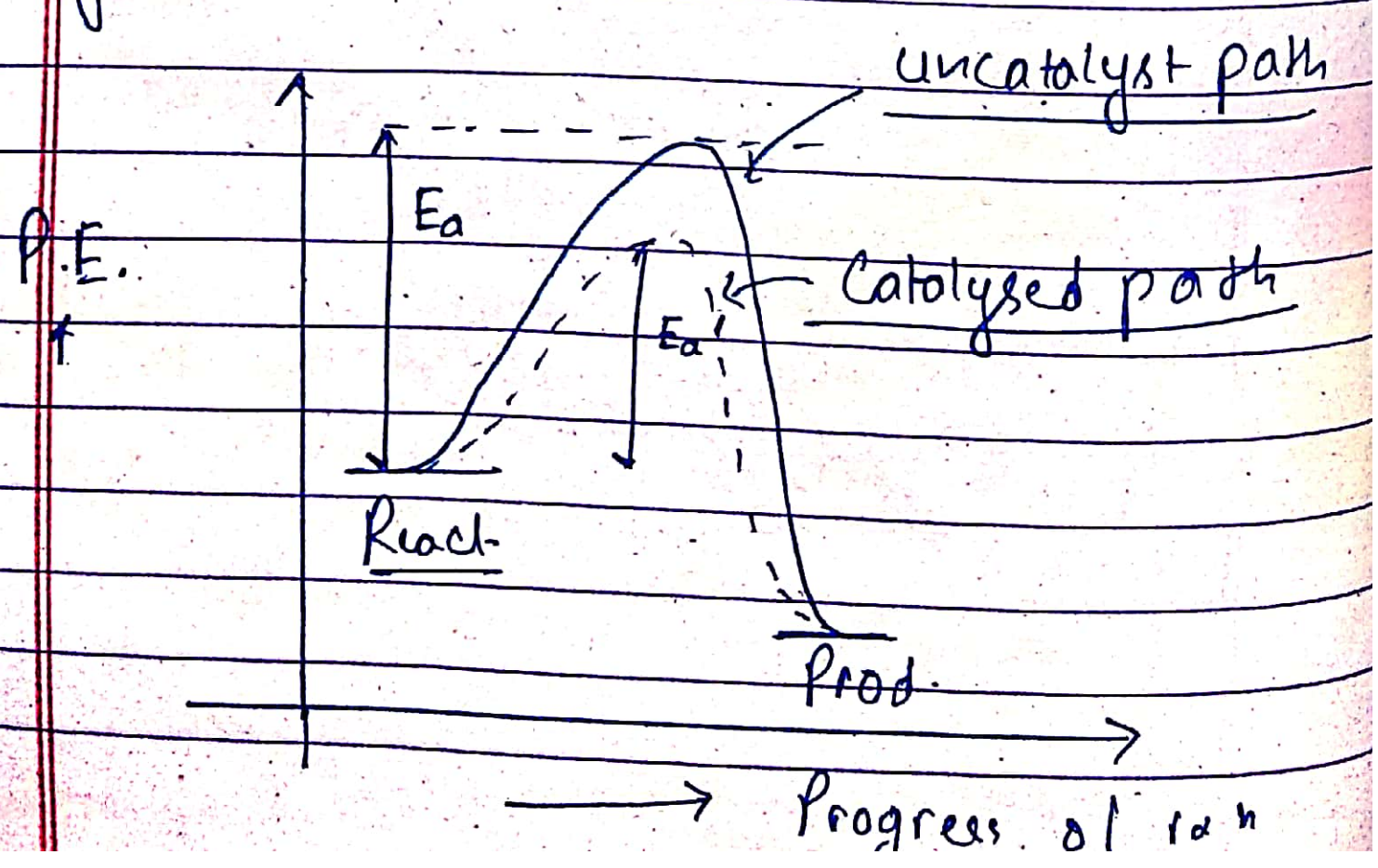
→ Catalyst is a substance which without being consumed increase the rate of chemical reaction.

→ Catalyst modifies the mechanism of reaction or it takes the reactants to products through an alternate path (involving lower activation energy (or low energy barrier)

eg.)

$$KClO_3\ (s) \xrightarrow[\text{Catalyst}]{H_2O_2} KCl\ (s) + 1.5\,O_2\ (g)$$

→ We can show the Potential energy diagram ⇒



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E_a → activation energy of uncatalysed path

E_a' → Activation energy of catalysed path

$(E_a' < E_a)$

→ Rate = K. (conc)$^{\text{order}}$

Rate ↑ if K ↑

& $K = Ae^{-E_a/RT}$ (for uncatalysed path)

$K' = Ae^{-E_a'/RT}$ (for catalysed path)

so, $\boxed{K' > K}$

If $K = K'$, then $\frac{E_a}{T} = \frac{E_a'}{T'}$ ⇒ useful & Imp

Since $E_a > E_a'$

$\therefore T' < T$

① Conclusion: Rxn can be carried out at lower temp. in presence of catalyst to maintain same rate

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V. Imp

→ • Catalyst doesn't initiate the reaction, i.e., non-spontaneous reaction cannot be turned into spontaneous ran by catalyst.

→ Catalyst doesn't effect Thermodynamic parameters of reaction like ΔH_{rxn}, ΔG_{rxn} etc.

→ Catalyst doesn't effect the value of eqlbm const. & position of eqlbm.

(*) catalyst decreases activation energy of both forward & backward direction by same value & hence, K_f & K_b (rate const.) increase by same factor, so their ratio ($\frac{K_f}{K_b}$ = K_{eq}) remains unchanged.

$$K_{eq} = \frac{K_f}{K_b} = \frac{A_f e^{-E_{af}/RT}}{A_b e^{-E_{ab}/RT}} = \frac{A_f}{A_b} e^{-(\Delta H_{rxn})/RT}$$

does not change by catalyst

(k) Negative Catalyst (or Inhibitor) ⇒

→ It decreases the rate of rxn, i.e., slows down the rxn.

→ One possible mechanism of activity of inhibitor is to consume one of the reactant involved in slow step (or rate determining step) of reaction mechanism, & hence, slows down the reaction.

*** There may be other mechanisms of inhibition of the rxn.**

(l) Effect of Surface Area on reaction ⇒

Imp → Higher is the surface area (i.e. in powdered form or in porous form), higher will be rate of rxn.

Quesⁿ In which case reaction of marble ($CaCO_3$) is slowest with HCl(aq)?

Ans. ⇒

(A) Marble Chips at 25°C

(B) Powdered marble at 20°C

(C) Marble Chips at 20°C } lower temp as well as not in powdered form

(D) Powdered marble at 25°C

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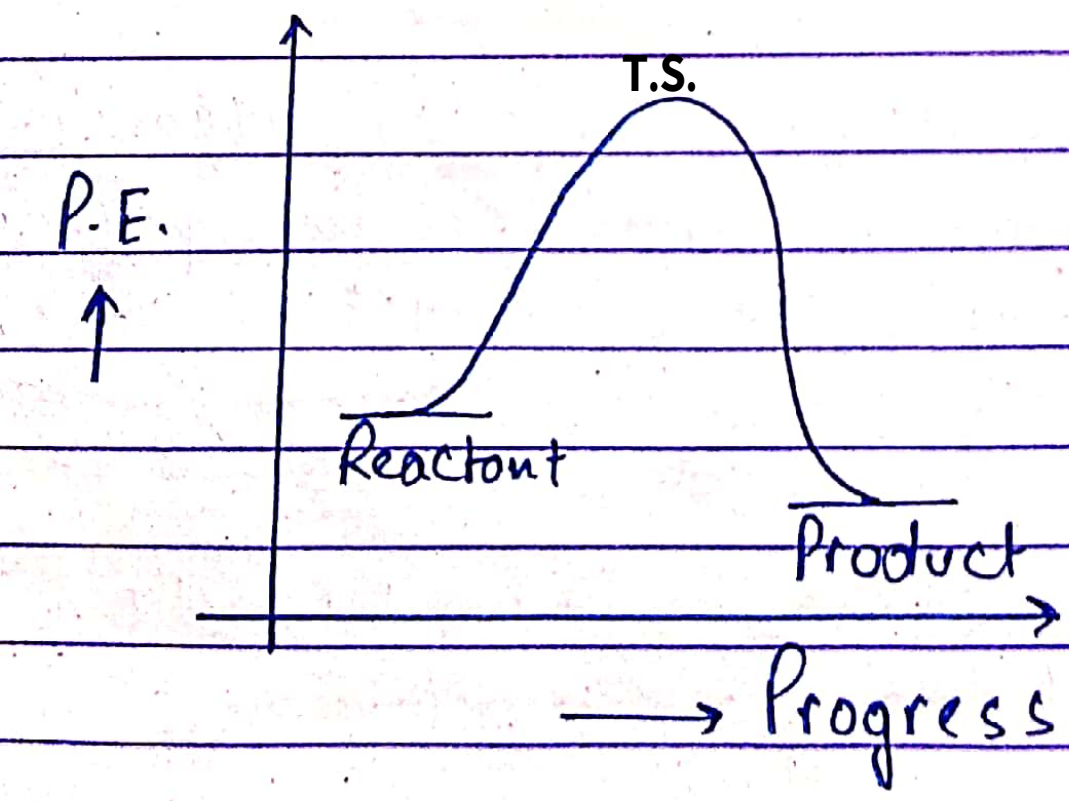
Reaction Mechanism ⟹

Type of Rxn → elementary reaction (or simple rxn)

→ complex reaction (or composite rxn)

(1) Elementary Reaction : Reaction which completes in single step or involving no experimentally detected intermediate.



eg.) $O_3 + NO \longrightarrow O_2 + NO_2$

[Elementary rxn]

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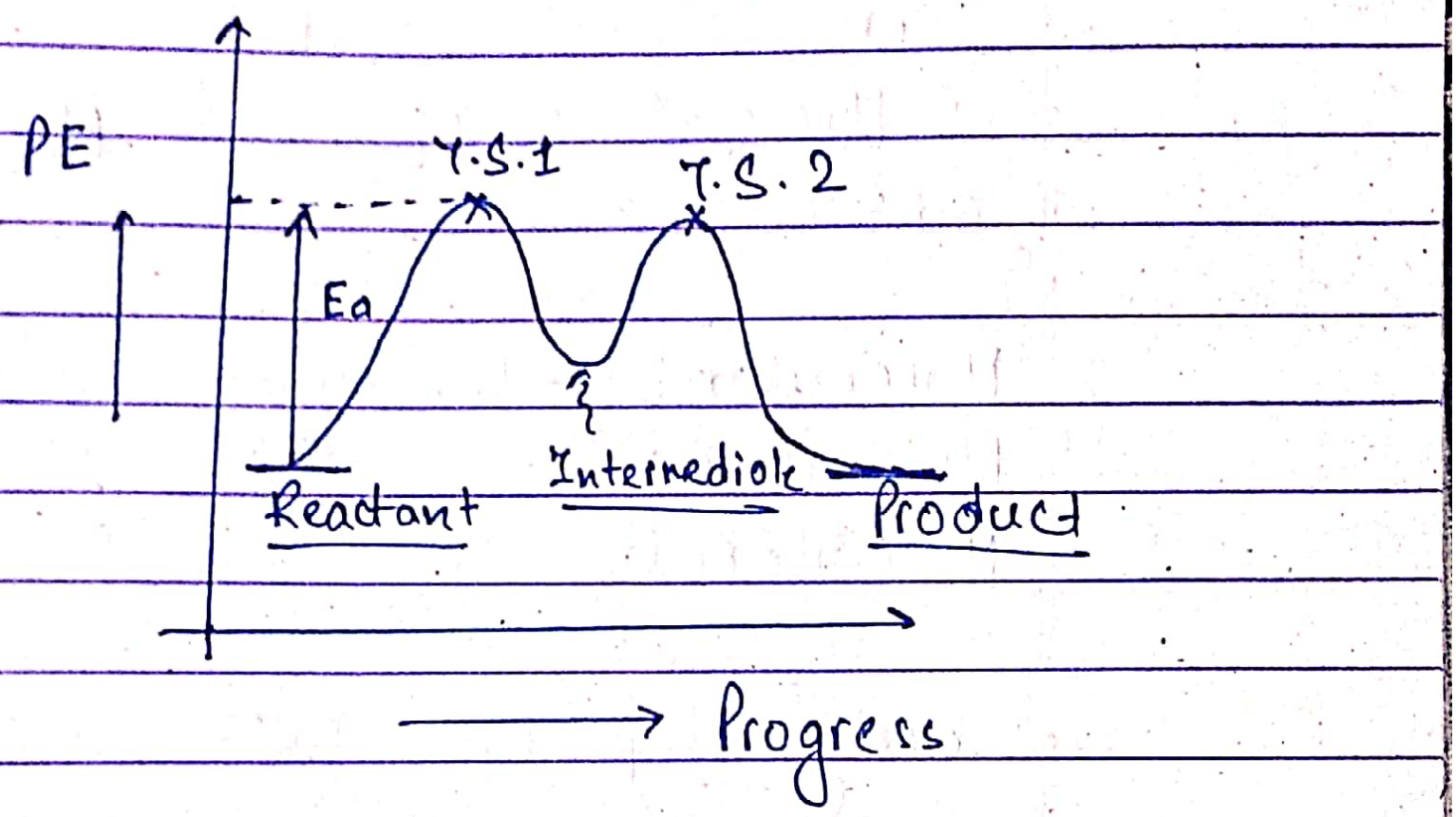
or Composite reaction

(2) Complex Reaction : Rxn which completes in two or more no. of steps. Each step is an elementary process.

It involves at least one intermediate



→ slowest step of reaction mechanism is called Rate Determining Step. (One which has the highest Ea)

↳ this step or elementary process determines the overall rate of reaction

Mechanism of Rxn ⇒ A collection of elementary processes for occurence of rxn is called Reaction Mechanism.

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Molecularity →

Molecularity of an element process (or elementary steps) is defined as the no. of reactants (molecule / ions / free radicals) which form transition state with sufficient energy & proper orientation.

Molecularity = 1, unimolecular rxn
Molecularity = 2, Bimolecular rxn
Molecularity = 3, Termolecular rxn

→ Every step of given mechanism is an elementary process.

(*) It is absurd or meaningless to define molecularity of composite rxn.

(+) Rather Molecularity is defined for every molecularity process

→ For an elementary process,

Molecularity = sum of its stoichometric coff of reactants

= order of elementary rxn

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HW

(b) A unimolecular elementary rxn is a first order rxn but its converse isn't true. [i.e. A first order rxn need not be a unimolecular elementary rxn]

Similarily, a bimolecular elementary rxn is a second order rxn, but converse isn't true.

(c) Molecularity can be 1, 2 & rarely 3, but not observed to be 4 or more because probability of 4 or more reactant molecules ~~extremely low~~. to collide is extremely low.

→ Molecularity cannot be fractional or zero

Therefore, a zero order rxn can't be an elementary rxn since molecularity cannot be zero

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(i) Examples of unimolecular process → (decomposition or rearrangement)

i) $O_3^* \longrightarrow O_2 + O$

(O_3^* = energised molecule) — unstable — [unimolecular]

* from energised molecule

ii) CH_2-CH_2 (cyclopropane) $\longrightarrow CH_3-CH=CH_2$ (propylene)

(energised molecule) — [unimolecular]

(k) Examples of Bimolecular process ⇒

eg.) $O_3 + Ar \longrightarrow O_3^* + Ar$

Molecularity = 2

(O_3^* = energised molecule) and becomes unstable

$O_3^* \longrightarrow O_2 + O$

(unimolecular step, Molecularity = 1)

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(e.g.) $NO + O_3 \longrightarrow NO_2 + O_2$

[Molecularity = 2]

(e.g.) $Cl^{\cdot} + CH_3^{\cdot} \longrightarrow HCl + CH_3Cl$

[Molecularity = 2]

Determination of Rate law from given Mechanism ⇒

→ Most important step for finding rate law is Rds (rate determining step) which is the slowest step in the given rxn.

CASE-1 : Elementary rxn is given (rds)

$NO + O_3 \xrightarrow{K} NO + O_2$ (rds)

Rate = $K[NO][O_3]$,

|| order = 2 ||

= molecularity

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Case - II : Complex rxn & rds involves reactant / products / catalysts in reactants but not intermediates.

(1) Rxn : $2NO_2 + F_2 \longrightarrow 2NO_2F$

Mechanism : $NO_2 + F_2 \xrightarrow{K_1} NO_2F + \underline{F}$ (slow)

rds

$\underline{F} + NO_2 \xrightarrow{K_2} NO_2F$ (fast)

- final rate cannot involve intermediate.

$$\text{Rate} = -\frac{d[NO_2]}{dt} = -\frac{d[F_2]}{dt}$$

$$= K_1[NO_2][F_2]$$

∴ second order

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Case-III : Complex rxn & rds involves intermediate in reactant & intermediate is in rapid eqlbm with reactants.

eg.)

Rxn : $2NO + O_2 \longrightarrow 2NO_2$

Mechanism : $NO + NO \underset{K_2}{\overset{K_1}{\rightleftharpoons}} N_2O_2$ [fast]

$N_2O_2 + O_2 \xrightarrow{K_3} 2NO_2$ [slow]

Sol^n :

$$Rate = K_3 [N_2O_2][O_2] \quad -(1)$$

↑ intermediates which aren't allowed

$$K_{eq} = \frac{K_1}{K_2} = \frac{[N_2O_2]}{[NO]^2}$$

$$\Rightarrow [N_2O_2] = \frac{K_1 [NO]^2}{K_2} \quad -(2)$$

order = 1+1+1 = 3

(2) in eqn (1) →

$$Rate = K_3 \times \frac{K_1}{K_2} [NO]^2 \times [O_2]$$

Final Rate law ⇒

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Case-4 : When information of fast, slow step isn't given in composite rxn.

(STEADY STATE APPROXIMATION)

→ In SSA, we assume all elementary steps (or processes) proceeding w/ same rate.

→ We assume conc. of intermediate to be constant during most reaction time period, except, in the beginning and at end.

Intermediate → I

$\frac{d[I]}{dt} = 0$ as rate is const

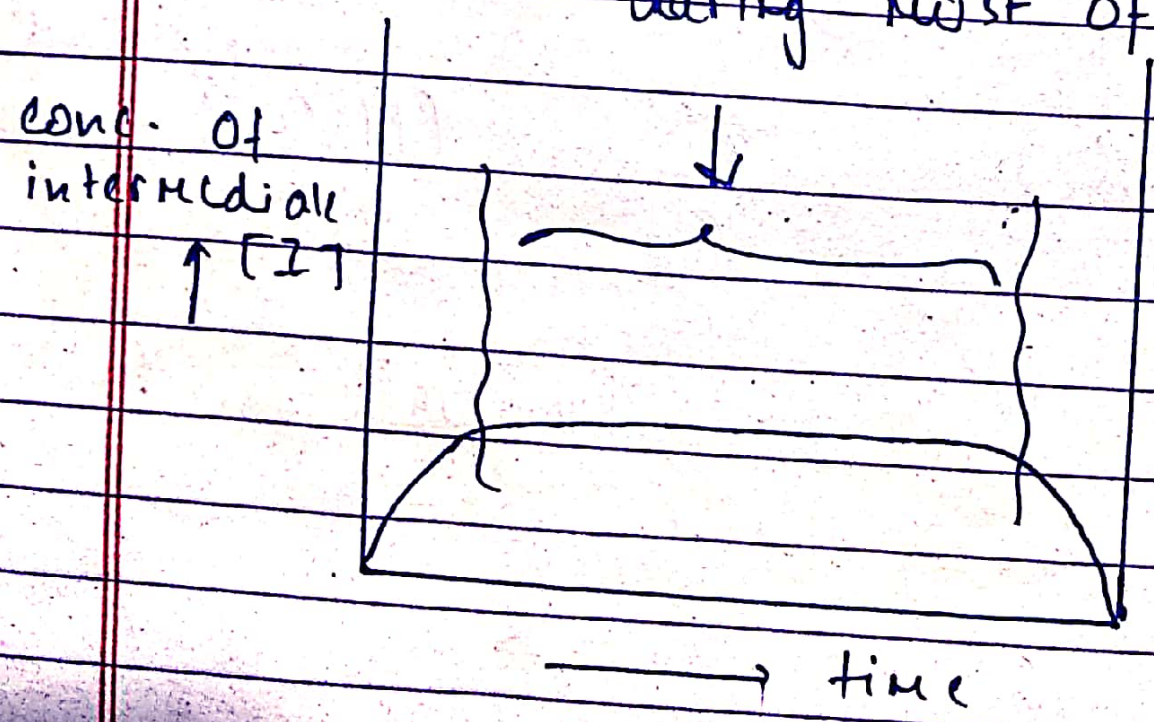
during most of rxn time period,

[I] = Const.

So,

$\frac{d[I]}{dt} = 0$

(SSA)



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eg.)

Rxn : $2N_2O_5 \longrightarrow 4NO_2 + O_2$

Mechanism : $N_2O_5 \underset{K_H}{\overset{K_1}{\rightleftharpoons}} NO_2 + NO_3$

$NO_3 + NO_2 \xrightarrow{K_2} NO + NO_2 + O_2$

$NO_3 + NO \xrightarrow{K_3} 2NO_2$

Soln :

$$\text{Rate} = \frac{d[O_2]}{dt} = K_2 \underbrace{[NO_3]}_{\downarrow I}[NO_2] \quad \text{— (1)}$$

Applying SSA on NO_3 ⇒

$$\frac{d[NO_3]}{dt} = 0$$

$$\Rightarrow K_1[N_2O_5] - K_H[NO_2][NO_3] - K_2[NO_3][NO_2] - K_3[NO_3][NO] = 0 \quad \text{— (2)}$$

Applying SSA on NO ⇒

$$\frac{d[NO]}{dt} = 0$$

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$\Rightarrow K_2[NO_3][NO_2] - \underbrace{K_3[NO_3][NO]} = 0$

↓

put this value in eq" (2)

$\Rightarrow$

$K_1[N_2O_5] - (K_4 + 2K_2)[NO_3][NO_2] = 0$

$$[NO_3] = \frac{K_1[N_2O_5]}{(K_4 + 2K_2)[NO_2]}$$

Put this value in eq" (1)

$\Rightarrow$

$$\text{Rate} = \frac{d[O_2]}{dt}$$

$$= K_2 \times \frac{K_1[N_2O_5]}{(K_4 + 2K_2)[NO_2]} \times [NO_2]$$

$$\Rightarrow \text{Rate} \Rightarrow \frac{K_2 K_1 [N_2O_5]}{(K_4 + 2K_2)}$$

order of the rex. = 1

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Q. Photochemical rxn →

$$2C_4H_{10} + h\nu \xrightarrow{\text{light}} C_{28}H_{20}$$

Mechanism →

$$C_4H_{10} + h\nu \xrightarrow[\text{intensity of light used}]{I} C_4H_{10}^*$$

(energised molecule)

$$C_4H_{10}^* + C_4H_{10} \xrightarrow{k_2} C_{28}H_{20}$$

$$C_4H_{10}^* \xrightarrow{k_3} C_4H_{10} + h\nu'$$

Prove that → $\frac{d[C_{28}H_{20}]}{dt} = \frac{I \times [C_4H_{10}]}{\frac{k_3}{k_2} + [C_4H_{10}]}$

(*) Remember → In photochemical reactions, rate of rxn depends on intensity of light ($h\nu$) used & not on conc. of reactants.

Soln : $\frac{d[C_{28}H_{20}]}{dt} = k_2[C_4H_{10}^*][C_4H_{10}]$

SSA on $C_4H_{10}^*$

$\frac{d[C_4H_{10}^*]}{dt} = 0$

$I - k_2[C_4H_{10}^*][C_4H_{10}] - k_3[C_4H_{10}^*] = 0$

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$$\Rightarrow K_2[C_{14}H_{10}] \times \frac{I}{K_3 + K_2[C_{14}H_{10}]}$$

* Rate $\Rightarrow$ $\dfrac{I\,[C_{14}H_{10}]}{\dfrac{K_3}{K_2} + [C_{14}H_{10}]}$

NOTE : for photochemical rx^{n}s, rxn rate depends on intensity of light used.

Q → Rxn : $H_2 + Br_2 \longrightarrow 2HBr$

Mechanism : (1) $Br_2 \xrightarrow{K_1} \underline{2Br}$

(2) $\underline{Br} + H_2 \xrightarrow{K_2} HBr + \underline{H}$

(3) $\underline{H} + Br_2 \xrightarrow{K_3} HBr + \underline{Br}$

(4) $\underline{H} + HBr \xrightarrow{K_4} H_2 + \underline{Br}$

(5) $\underline{Br} + \underline{Br} \xrightarrow{K_5} Br_2$

Prove that,

$$\frac{d[HBr]}{dt} = \frac{2K_2\left(\frac{K_1}{K_5}\right)^{\frac{1}{2}}[H_2][Br_2]}{1 + \frac{K_4}{K_3}\frac{[HBr]}{[Br_2]}}$$

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Soln:

$$Rate = \frac{d[HBr]}{dt} = K_2[Br][H_2] + K_3[H][Br_2] - K_4[H][HBr] \quad \text{— (1)}$$

Applying SSA on Br atom ⇒

$$\frac{d[Br]}{dt} = 0$$

$$2K_1[Br_2] - K_2[Br][H_2] + K_3[H][Br_2] + K_4[H][HBr] - 2K_5[Br]^2 = 0 \quad \text{— (2)}$$

Applying SSA on H atom ⇒

$$\frac{d[H]}{dt} = 0$$

$$K_2[Br][H_2] - K_3[H][Br_2] - K_4[H][HBr] = 0 \quad \text{— (3)}$$

Put (3) in (2) →

$$2K_1[Br_2] - 2K_5[Br]^2 = 0$$

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$$[Br] = \left\{ \frac{2K_1}{2K_5} [Br_2] \right\}^{\frac{1}{2}}$$

$$\left\{ [Br] \Rightarrow \left(\frac{K_1}{K_5}\right)^{\frac{1}{2}} [Br_2]^{\frac{1}{2}} \right. \quad \leftarrow (4)$$

$\downarrow$ Put this in eqn (3)

$$\Rightarrow [HT] = K_2[H_2] \times \left(\frac{K_1}{K_5}\right)^{\frac{1}{2}} [Br_2]^{\frac{1}{2}}$$

$$\overline{K_3[Br_2] + K_4[HBr]} \quad - (5)$$

Putting eqn (4) & (5) in eqn (1) $\rightarrow$

$$\frac{d[HBr]}{dt} = K_2[Br][H_2] + [H] \left\{ K_3[Br_2] - K_4[HBr] \right.$$

$$\Rightarrow K_2[H_2]\left(\frac{K_1}{K_5}\right)^{\frac{1}{2}} [Br_2]^{\frac{1}{2}} +$$

$$\left(\frac{K_2[H_2] \times \left(\frac{K_1}{K_5}\right)^{\frac{1}{2}} [Br_2]^{\frac{1}{2}}}{K_3[Br_2] + K_4[HBr]} \right)$$

$$\cdot \left(K_3[Br_2] - K_4[HBr] \right.$$

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$$\text{Rate} = \frac{d[HBr]}{dt}$$

$$\Rightarrow K_2[H_2]\left(\frac{K_1}{K_5}\right)^{\frac{1}{2}}[Br_2]^{\frac{1}{2}} \times$$

$$\left(1 + \frac{K_3[Br_2] - K_4[HBr]}{K_3[Br_2] + K_4[HBr}\right)$$

$$\Rightarrow K_2[H_2]\left(\frac{K_1}{K_5}\right)^{\frac{1}{2}}[Br_2]^{\frac{1}{2}}\left(\frac{2K_3[Br_2]}{(K_3[Br_2] + K_4[HBr])}\right)$$

$$\Rightarrow \frac{2K_2\left(\frac{K_1}{K_5}\right)^{\frac{1}{2}}[H_2][Br_2]^{\frac{1}{2}}}{1 + \frac{K_4[HBr]}{K_3[Br_2]}}$$

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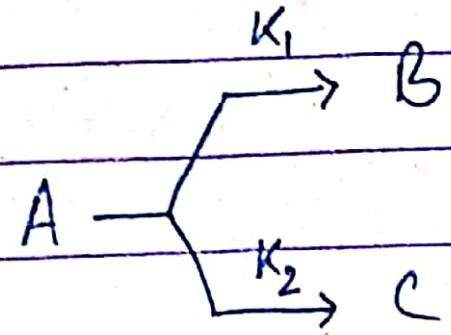
Some complicated first order reactions ⇒

(1) Parallel or competing reactions ⇒

First Order



B & C of

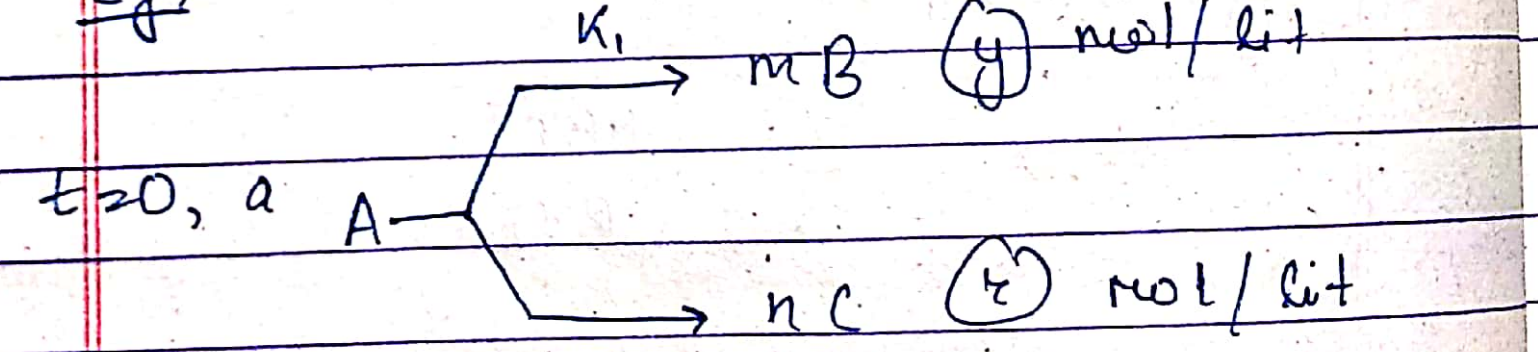
Normal reaction ⇒

$A \xrightarrow{K} B + C$

1 : 1 molar ratio

Amount of such competing reactions depend on K_1 & K_2 value.

eg:



$\left.\begin{array}{l} \because \text{ m mol of B} \rightarrow \text{1 mol of A} \\ \text{y mol of B} \rightarrow \frac{1}{m} \times y \text{ mol of A} \end{array}\right\}$

Similarity,

$\left.\begin{array}{l} \text{z mol of C} \rightarrow \frac{1}{n} \times z \text{ mol of A} \end{array}\right\}$

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Total conc. used of A = x

$$\rightarrow x = \frac{y}{m} + \frac{z}{n} \quad \text{— (1)}$$

$$\Rightarrow \frac{dx}{dt} = \frac{1}{m}\frac{dy}{dt} + \frac{1}{n}\frac{dz}{dt}$$

$$\Rightarrow -\frac{d[A]}{dt} = \frac{1}{m}\frac{d[B]}{dt} + \frac{1}{n}\frac{d[C]}{dt}$$

Since, $A \xrightarrow{K_1} mB$, $A \xrightarrow{K_2} nC$

$$\text{So, } -\frac{d[A]}{dt} = \frac{1}{m} \times m K_1 [A]_t + \frac{1}{n} \times n K_2 [A]_t$$

$$\Rightarrow -\frac{d[A]}{dt} = (K_1 + K_2)[A]_t$$

$$\Rightarrow -\int_{[A]_0}^{[A]_t} \frac{d[A]}{[A]_t} = \int_0^t (K_1 + K_2)\, dt$$

$$\Rightarrow \boxed{[A]_t = [A]_0 e^{-(K_1 + K_2)t}}$$

$\Rightarrow$ for parallel 1^{st} order, effective rate const. of 'n'

$$K = K_1 + K_2$$

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Half life ⇒

$$\frac{\ln 2}{(t_{1/2})_{overall}} = \frac{\ln 2}{(t_{1/2})_1} + \frac{\ln 2}{(t_{1/2})_2}$$

Using Arrhenius eqⁿ,

$$Ae^{-E_0/RT} = A_1 e^{-E_0/RT} + A_2 e^{-E_{a_2}/R}$$

Diff. wrt time ⇒

$$Ae^{-E_0/RT} \times \left(\frac{E_a}{RT^2}\right) = A_1 e^{-E_{a_1}/RT} \left(\frac{E_{a_1}}{RT^2}\right) + A_2 e^{-E_{a_2}/RT} \left(\frac{E_{a_2}}{RT^2}\right)$$

$Ae^{-E_0/RT}$ → K

$A_1 e^{-E_{a_1}/RT}$ → K_1

$A_2 e^{-E_{a_2}/RT}$ → K_2

$$\Rightarrow K \times E_a = K_1 \times E_{a_1} + K_2 \times E_{a_2}$$

$$\Rightarrow E_a = \frac{K_1 E_{a_1} + K_2 E_{a_2}}{K = k1 + k2}$$

Overall activation energy of parallel rxn of type 'A'

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Conc. of B & C ⇒

$$\frac{d[B]_t}{dt} = mK_1[A]_t = mK_1[A]_0 e^{-(K_1+K_2)t}$$

$$\Rightarrow \int_0^{[B]_t} d[B]_t = mK_1[A]_0 \int_0^t e^{-(K_1+K_2)t}\, dt$$

$$\Rightarrow [B]_t = \frac{mK_1[A]_0}{-(K_1+K_2)} \times \left\{ e^{-(K_1+K_2)t} \right\}_0^t$$

$$\Rightarrow \boxed{[B]_t = \frac{mK_1[A]_0}{(K_1+K_2)} \left(1 - e^{-(K_1+K_2)t} \right)}$$

Similarly,

$$\boxed{[C]_t = \frac{nK_2[A]_0}{(K_1+K_2)} \left(1 - e^{-(K_1+K_2)t} \right)}$$

And, $\frac{[B]_t}{[C]_t} = \frac{mK_1}{nK_2}$ //

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Q: $A \xrightarrow{K_1} B$, $A \xrightarrow{K_2} C$ $\quad K_1 = 0.1\ hr^{-1}$

$\frac{K_1}{K_2} = \frac{1}{10}$

Find $\frac{[C]}{[A]}$ after 1hr from start of reaction. Assume reaction started w/ only A & both are 1st order rxn.

Soln

$K_1 = 0.1\ hr^{-1}$

$\Rightarrow K_2 = 1\ hr^{-1}$

$$\frac{[C]}{[A]} = \frac{\frac{1 \times K_2 \times [A]_0}{K_1 + K_2}\left\{1 - e^{-(K_1+K_2)t}\right\}}{[A]_0\, e^{-(K_1+K_2)t}}$$

$$\Rightarrow \frac{1}{1.1}\left(e^{1.1t} - 1\right)$$

$$\Rightarrow \frac{1}{1.1}\left(e^{1.1\times 1} - 1\right)$$

$$\frac{[C]}{[A]} \Rightarrow \frac{1}{1.1}\left(e^{1.1} - 1\right)$$

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[JEE - ADVANCE]

Imp

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Ques Consider a first order parallel rxn

$$A(g) \xrightarrow{K_1} 2B(g)$$
$$A(g) \xrightarrow{K_2} C(g)$$

In initial pressure of A in a container of fixed vol is 1atm. Pressure of system at $t = 10\,s$ is 1.4 atm & after ∞ time, it becomes 1.5 atm. Find partial pressure of A at $t = 10\,sec$ & K_1 and K_2.

Sol^n:

$$A \xrightarrow{K_1} 2B(g) \quad | \quad t = 10\,s,\ P_T = 1.4\,atm$$
$$A \xrightarrow{K_2} C(g) \quad | \quad t = \infty,\ P_T = 1.5\,atm$$

At $t = \infty$, A gets exhausted fully, x press. of A goes to be B press. & $(1-x)$ goes to C.

So, At $t = \infty$ → 2B, C

$P_B = 2x$, $P_C = (1-x)$

At $t = \infty$ $P_T = P_A + P_B + P_C$

$1.5 = 0 + 2x + 1 - x$

$x = 0.5$

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Now,

$$\left(\frac{P_B}{P_C}\right)_\infty = \frac{[B]_\infty}{[C]} = \frac{2K_1}{1K_2}$$

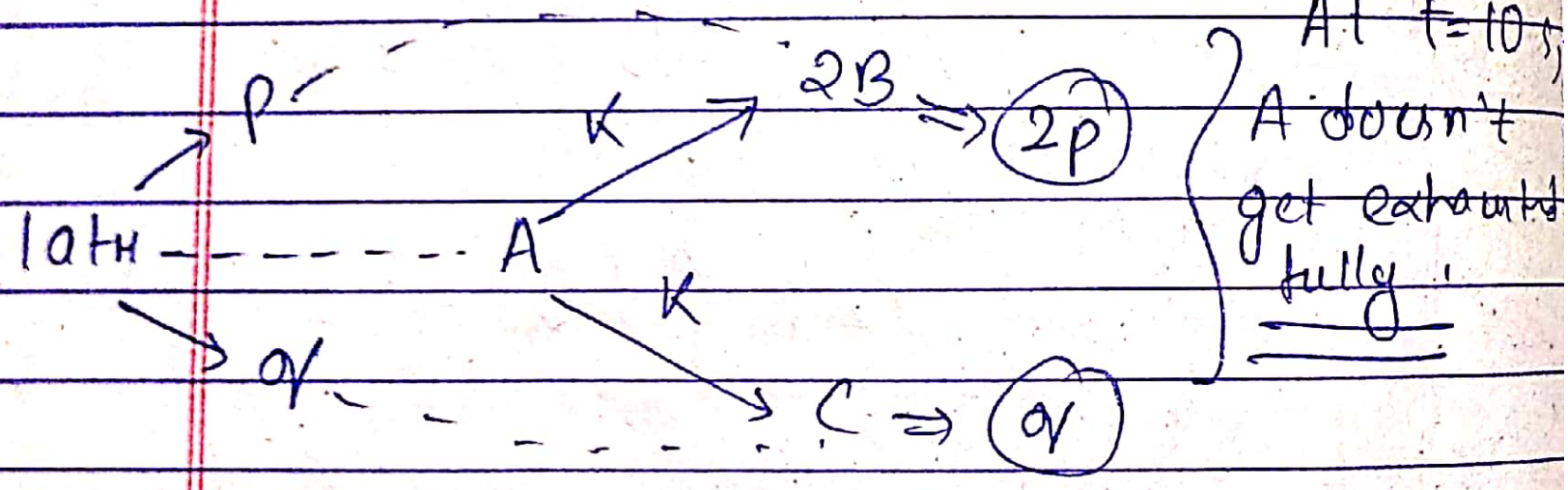
$$= \frac{2 \times 0.5}{1 - 0.5}$$

$$\Rightarrow \frac{2K_1}{K_2} = \frac{1}{0.5}$$

$$\Rightarrow \frac{2K_1}{K_2} = 2$$

$$\Rightarrow K_1 = K_2 \Rightarrow K$$

At $t = 10s \rightarrow$



At t=10s, A doesn't get exhausted fully.

$$2p + Q + (1 - p - Q) = 1.4$$

$$p + 1 = 0.4$$

$$p = 0.4 \text{ atm}$$

$$\left(\frac{P_B}{P_C}\right)_{t=10s} = \frac{2p}{Q} = \frac{2K}{1K}$$

$$\Rightarrow p = Q = 0.4 \text{ atm}$$

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i) $(P_A)_{t=10s} = 1 - P - q$

$= 1 - 0.4 - 0.4$

$= 0.2 \text{ atm}$

$[A]_t = [A]_0 \, e^{-(K_1 + K_2)t}$

$\Rightarrow 0.2 = e^{-2K \times 10}$

$\Rightarrow \ln(0.2) = -20K$

$\Rightarrow K = \frac{0.7 \times 2.3}{20}$

≈ 0.08

ii) $\{K_1 = K_2 = 0.08\}$

(2) Reversible Reactions $\Rightarrow$

(1st Order)

$t=0$,	a	0
	$A \underset{K_2}{\overset{K_1}{\rightleftharpoons}}$	B
$t=t$,	$a-x$	x
$t=\infty$,	$a-x_e$	x_e

In reversible rxns, conc. of reactant can never be zero,

$x_e \rightarrow$ is the conc. of A used up at eqlbm state.

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$$\Rightarrow K_{eq} = \frac{K_1}{K_2} = \frac{[B]_{eq}}{[A]_{eq}} = \frac{x_e}{a - x_e}$$

$$\Rightarrow a K_1 = (K_1 + K_2) x_e \quad \text{— (1)}$$

$\Rightarrow$ Rate of Disappearence of A $\rightarrow$

$$-\frac{d[A]}{dt} = 1 \times K_1 [A]_t - 1 K_2 [B]_t$$

$$\Rightarrow -\frac{d}{dt}(a - x) = 1 \times K_1 (a - x) - K_2 (x)$$

↳ using eq^n (1)

$$\frac{dx}{dt} = K_1 a - (K_1 + K_2) x$$

$$\Rightarrow \frac{dx}{dt} = (K_1 + K_2)(x_e - x)$$

$$\Rightarrow \int_0^x \frac{dx}{(x_e - x)} = (K_1 + K_2) \int_0^t dt$$

$$\Rightarrow \{-\ln(x_e - x)\}_0^x = (K_1 + K_2)\, t$$

$$\Rightarrow \boxed{t = \frac{1}{(K_1 + K_2)} \cdot \ln\left(\frac{x_e}{x_e - x}\right)}$$

Remember

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Q: For a reversible 1st order rxn →

$$A \underset{K_2}{\overset{K_1 = 10^{-2}\ sec^{-1}}{\rightleftharpoons}} B$$

$\frac{[B]}{[A]}$ at eqⁿ = 4

If $[A]_0 = 10^{-2}$ M & $[B]_0 = 0$, find conc. of B after 30s.

$e^{0.375} = 1.5$ J

Soln:

$$A \underset{K_2}{\overset{K_1}{\rightleftharpoons}} B$$

$K_1 = 10^{-2}\ sec^{-1}$

$K_{eq} = \frac{K_1}{K_2} = 4$

$\frac{K_1}{K_2} = 4 = \frac{x_e}{10^{-2} - x_e}$

$\Rightarrow 4 \times 10^{-2} = 5 x_e$

$\Rightarrow x_e = 0.8 \times 10^{-2}$ M

$$\Rightarrow 30 = \frac{1}{10^{-2} + \frac{10^{-2}}{4}} \ln\left(\frac{8 \times 10^{-3}}{8 \times 10^{-3} - x}\right)$$

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$$30 = \frac{4}{5\times10^{-2}} = \ln\left(\frac{8\times10^{-3}}{8\times10^{-3}-x}\right)$$

$$\Rightarrow 30\times1.25\times10^{-2} = \ln\left(\frac{8\times10^{-3}}{8\times10^{-3}-x}\right)$$

$$\Rightarrow 0.375 = \ln\left(\frac{8\times10^{-3}}{8\times10^{-3}-x}\right)$$

$$\Rightarrow \frac{8\times10^{-3}}{8\times10^{-3}-x} = e^{0.375} = 1.5$$

$$\Rightarrow 8\times10^{-3} = 12\times10^{-3} - \frac{3x}{2}$$

$$\Rightarrow \frac{3x}{2} = 4\times10^{-3}$$

$$\Rightarrow x = \frac{8}{3}\times10^{-3}\ M$$

$$[B]_{30sec} = \quad\quad = 1$$

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(3) First Order Consequtive or Sequencial Rxn ⟹

(observed in Radioactive decay)

$$A \xrightarrow{} Y \xrightarrow{} P$$

(Reactant) (Intermediate) (Product)

$t=0$,	a	0	0
$t=t$,	$a-x$	$x-y$	y

$$-\frac{d[A]_t}{dt} = K_1 \times [A]_t$$

⇒ after integrating :

$$\boxed{[A]_t = [A]_0 e^{-kt}}$$

$$+\frac{d[I]_t}{dt} = K_1[A] - K_2[I]$$

$$\Rightarrow \left(\frac{d[I]_t}{dt} + K_2[I]\right) \cdot e^{K_2 t} = K_1[A] \cdot e^{K_2 t}$$

$$\Rightarrow \frac{d}{dt}\left\{e^{K_2 t} \times [I]\right\} = K_1[A]_0 e^{(K_2-K_1)t}$$

$$\Rightarrow d\left(e^{K_2 t} \times [I]_t\right) = K_1[A]_0 \int_0^t e^{(K_2-K)t} \cdot dt$$

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$$e^{K_2 t} \times [Y]_t = \frac{K_1 [A]_0}{(K_2 - K_1)} \left[e^{(K_2 - K_1)t} - 1 \right]$$

$$\Rightarrow [Y]_t = \frac{K_1 [A]_0}{(K_2 - K_1)} \left[e^{-K_1 t} - e^{-K_2 t} \right]$$

For maxima of $[Y]_t \rightarrow$

$$\frac{d[Y]_t}{dt} = 0$$

$$-K_1 e^{-K_1 t_{max}} + K_2 e^{-K_2 t_{max}} = 0$$

$$\Rightarrow K_1 e^{-K_1 t_{max}} = K_2 e^{-K_2 t_{max}}$$

$$\Rightarrow e^{(K_2 - K_1) t_{max}} = \frac{K_2}{K_1}$$

$$\Rightarrow (K_2 - K_1) t_{max} = \ln\left(\frac{K_2}{K_1}\right)$$

$$\Rightarrow \boxed{t_{max} = \frac{\ln(K_2/K_1)}{(K_2 - K_1)}}$$

⇓

Time from start at which conc. of intermediate reaches maximum (as it is 0 at start & 0 at end)

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Calculation of conc. of $[P]_t$:

~~$[A]_t + [Y]_t$~~

$$A \xrightarrow{K_1} Y \xrightarrow{K_2} P$$

$t=0$	$a-x$	$x-y$	y

$$[A]_t + [Y]_t + [P]_t = a - x + x - y + y$$

$$= a = [A]_0$$

$$[P]_t = [A]_0 - [A]_0 e^{-K_1 t} - \frac{K_1[A]_0}{(K_2 - K_1)}\left(e^{-K_1 t} - e^{-K_2 t}\right)$$

$$\Rightarrow \frac{[A]_0}{(K_2 - K_1)}\left\{K_2 - K_1 - K_2 e^{-K_1 t} + K_1 e^{-K_1 t} - K_1 e^{-K_2 t} + K_1 e^{-K_2 t}\right\}$$

$$[P]_t = \frac{[A]_0}{(K_2 - K_1)}\left\{K_2\left(1 - e^{-K_1 t}\right) - K_1\left(1 - e^{-K_2 t}\right)\right\}$$

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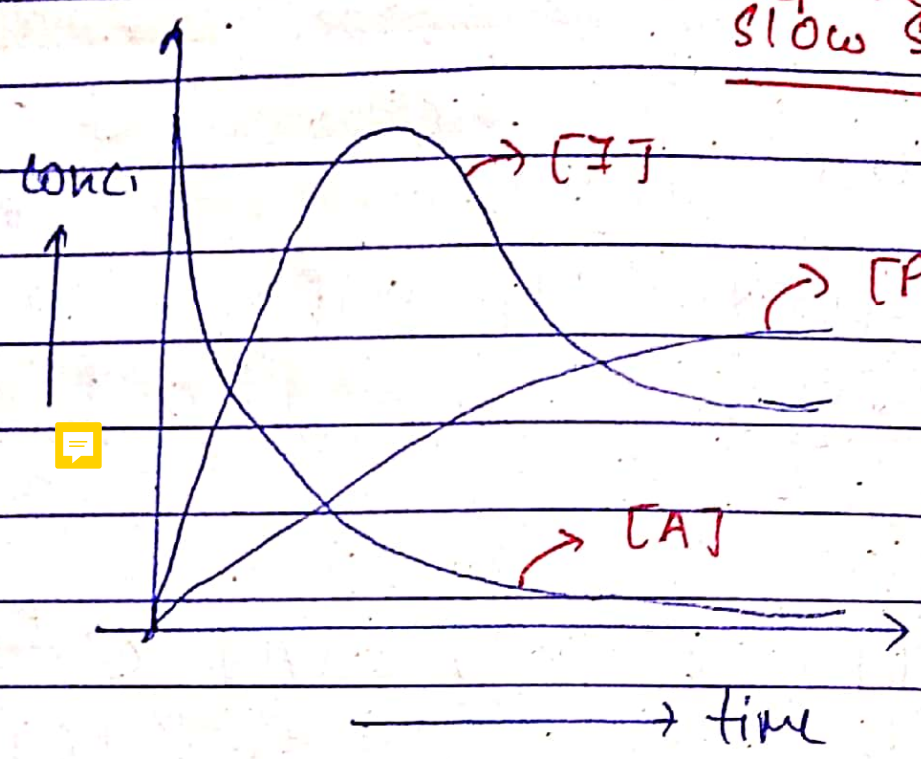
Case-I : if $K_1 >> K_2$,

$$A \xrightarrow{K_1} \underline{I} \xrightarrow{K_2} P$$

↳ very imp step as $K_1 >> K_2$ so, this step becomes the slow step (rds)

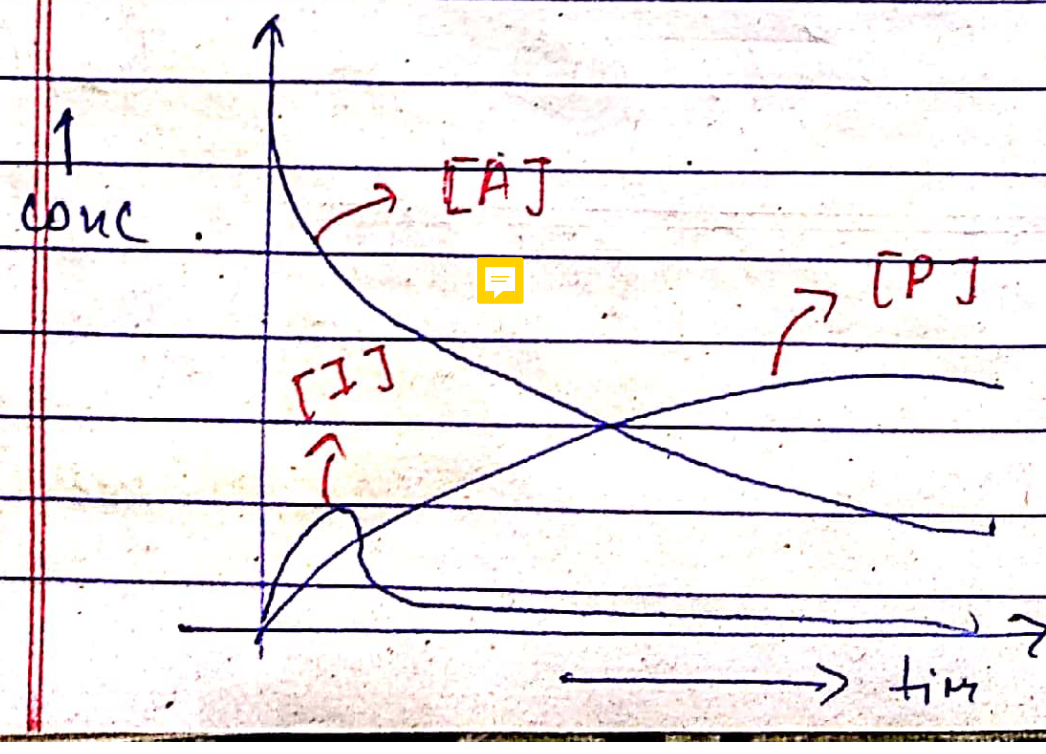


Case-II : if $K_2 >> K_1$,

$$A \xrightarrow{K_1} I \xrightarrow{K_2} P$$

↓

Imp



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